THE FINITE DEGREE FORMULA FOR F-VOLUMES

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Abstract. We prove that the F-volume and the Hilbert–Kunz F-volume of strongly F-regular singularities are multiplied by the degree under finite morphisms that are étale in codimension one. We thereby solve Problem 49 in Xu and Zhuang's list of open problems on K-stability of Fano varieties. The pair case also holds.

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1. Introduction

We work over an algebraically closed field k of characteristic $p > 0$.

For a klt singularity $(X \ni x)$ over $\mathbb{C}$ with $\dim X = d$, the normalized volume $\widehat{\mathrm{vol}}(X \ni x)$ is the infimum of the normalized volumes $\widehat{\mathrm{vol}}(v)$ of the valuations v centered at x [Li15, Section 4, (19)]. By [Liu18, Theorem 27],

$$\widehat{\mathrm{vol}}(X \ni x) = \inf_{\mathfrak{a}} e(\mathfrak{a}) \operatorname{lct}(\mathfrak{a})^d,$$

where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$, $\mathfrak{a}$ runs over the $\mathfrak{m}_x$-primary ideals of $\mathcal{O}_{X,x}$, $e(\mathfrak{a})$ is the Hilbert–Samuel multiplicity of $\mathfrak{a}$, and $\operatorname{lct}(\mathfrak{a})$ is the log canonical threshold of $\mathfrak{a}$. In positive characteristic, strongly F-regular singularities are analogous to klt singularities [HW02] (cf. [ST10, Section 6.1]), and the F-pure threshold [TW03, Definition 2.1] is related to the log canonical threshold through reduction modulo p (see [HY03, Theorems 3.4 and 6.8]). Let X be a normal variety with $\dim X = d \geq 1$ and $x \in X$ a closed point such that X is strongly F-regular at x. Replacing the log canonical threshold by the F-pure threshold in the formula of [Liu18, Theorem 27], one obtains the F-volume (see [LLX18, Definition 6.17]), and replacing moreover $e(\mathfrak{a})$ by the Hilbert–Kunz multiplicity $e_{\mathrm{HK}}(\mathfrak{a})$, one obtains the Hilbert–Kunz F-volume (see [XZ26, Section 4.3]):

$$\mathrm{Fvol}(X \ni x) := \inf_{\mathfrak{a}} e(\mathfrak{a}) \operatorname{fpt}(X \ni x; \mathfrak{a})^d, \qquad \mathrm{Fvol}_{\mathrm{HK}}(X \ni x) := \inf_{\mathfrak{a}} e_{\mathrm{HK}}(\mathfrak{a}) \operatorname{fpt}(X \ni x; \mathfrak{a})^d,$$

where $\mathfrak{a}$ runs over the $\mathfrak{m}_x$-primary ideals of $\mathcal{O}_{X,x}$ and $\operatorname{fpt}(X \ni x; \mathfrak{a})$ is the F-pure threshold of $\mathfrak{a}$ (see Definitions 2.2 and 2.4 for strong F-regularity, F-pure thresholds, and Hilbert–Kunz

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multiplicities). If X is smooth at x, then $\mathrm{Fvol}(X \ni x) = d^d$, and the infimum defining $\mathrm{Fvol}(X \ni x)$ is attained at the powers of $\mathfrak{m}_x$ [TW03, Proposition 4.5] (see [LLX18, Section 6.3.1]). Li, Liu, and Xu conjectured that the normalized volume of a klt singularity over $\mathbb{C}$ is the limit of the F-volumes of the reductions modulo p as $p \to +\infty$ [LLX18, Conjecture 6.19].

Xu and Zhuang asked the following question.

Question 1.1 ([XZ26, Problem 49])**.** Let $f : (Y \ni y) \to (X \ni x)$ be a finite morphism between strongly F-regular germs that is étale in codimension one. Is it true that

$$\mathrm{Fvol}(Y \ni y) = \deg(f) \cdot \mathrm{Fvol}(X \ni x) \quad \text{and} \quad \mathrm{Fvol}_{\mathrm{HK}}(Y \ni y) = \deg(f) \cdot \mathrm{Fvol}_{\mathrm{HK}}(X \ni x)?$$

Here a finite morphism $f : (Y \ni y) \to (X \ni x)$ between germs is a finite surjective morphism $f : Y \to X$ between normal varieties with $f^{-1}(x) = \{y\}$.

We now recall the analogous formula for the normalized volume, known as the finite degree formula. For a klt singularity $(X \ni x, B)$ over $\mathbb{C}$, we denote by $\widehat{\mathrm{vol}}(X \ni x, B)$ the normalized volume of $(X \ni x, B)$ (see [LLX18, Definition 2.3]). Let $f : (Y \ni y, B_Y) \to (X \ni x, B)$ be a finite morphism between klt singularities over $\mathbb{C}$ with $K_Y + B_Y = f^*(K_X + B)$. Liu and Xu conjectured that $\widehat{\mathrm{vol}}(Y \ni y) = \deg(f) \cdot \widehat{\mathrm{vol}}(X \ni x)$ if $B = 0$, f is the quotient by a finite group, and f is étale in codimension one [LX17, Conjecture 4.1]. Li, Liu, and Xu conjectured that

$$\widehat{\mathrm{vol}}(Y \ni y, B_Y) = \deg(f) \cdot \widehat{\mathrm{vol}}(X \ni x, B)$$

in general [LLX18, Conjecture 6.4]. The formula holds if f is Galois [XZ20, Theorem 1.3]. If f is étale in codimension one, then the formula reduces to the Galois case by passing to the Galois closure [XZ20, Section 4]. The general case is [Liu26, Theorem 1.1].

In positive characteristic, for a normal variety X and a closed point $x \in X$, the F-signature $s(X \ni x)$ of $\mathcal{O}_{X,x}$ (defined in [HL04, Definition 9] for Cohen–Macaulay rings and in [Tuc11, Theorem 4.9] in general) exists [Tuc11, Theorem 4.9], and $s(X \ni x) > 0$ if and only if X is strongly F-regular at x [AL02, Theorem 0.2]. If $f : (Y \ni y) \to (X \ni x)$ is a finite morphism that is étale in codimension one and X is strongly F-regular at x, then

$$s(Y \ni y) = \deg(f) \cdot s(X \ni x)$$

by [CST18, Theorem 3.1], and [CST18, Theorem 4.4] is a version for pairs. The formula was extended to F-signatures of Cartier algebras [CS19, Theorem 4.3] and to the perfectoid signature in mixed characteristic [Cai+22, Theorem 6.0.1].

We now explain why the methods of [CST18, Theorem 3.1] and [XZ20, Theorem 1.3] do not extend directly to the F-volume. In positive characteristic, [CST18, Theorem 3.1] is obtained by comparing, for any integer $e \geq 1$, the numbers of free summands of $\mathcal{O}_{Y,y}$, regarded as a module through the e-th iterate of the Frobenius, over $\mathcal{O}_{X,x}$ and over $\mathcal{O}_{Y,y}$, by means of the trace map. [JS19, Theorem 2.10] extends the method of [CST18, Theorem 3.1] to invariants of local rings defined asymptotically from the lengths $\ell_R(R/I_n(R))$, where Λ is an infinite set of positive integers and each local ring R in a given class, with maximal ideal $\mathfrak{m}_R$, is assigned a family $(I_n(R))_{n\in\Lambda}$ of $\mathfrak{m}_R$-primary ideals. The key condition is that $I_n(S) \cap R = I_n(R)$ for any $n \in \Lambda$ and any finite inclusion $R \subset S$ of local domains that is split and étale in codimension one. The F-volume is not defined from such a family of ideals: $\mathrm{Fvol}(X \ni x)$ is the infimum of $e(\mathfrak{a})\,\mathrm{fpt}(X \ni x; \mathfrak{a})^d$ over all the $\mathfrak{m}_x$-primary ideals $\mathfrak{a}$. Recall that if X is smooth at x, then the infimum is attained at the powers of $\mathfrak{m}_x$. However, the family $I_n(R) = \mathfrak{m}_R^n$ does not satisfy the key condition: if $p \neq 2$, then the inclusion $R = k[[u^2, uv, v^2]] \subset S = k[[u, v]]$ is finite, split, and étale in codimension one, but

$$\mathfrak{m}_S^2 \cap R = \mathfrak{m}_R \neq \mathfrak{m}_R^2$$

(see the paragraph after [JS19, Theorem 2.10]). Over $\mathbb{C}$, [XZ20, Theorem 1.3] relies on the uniqueness of the valuation computing the normalized volume. For instance, let the morphism $f : (Y \ni y) \to (X \ni x)$ be the quotient of a klt singularity by a faithful action of a finite group G such that f is étale in codimension one. There exists a valuation v_0 centered at y with

$\widehat{\mathrm{vol}}(v_0) = \widehat{\mathrm{vol}}(Y \ni y)$ [Blu16, Main Theorem], and v_0 is unique up to scaling [XZ20, Theorem 1.1]. Thus v_0 is G-invariant [XZ20, Corollary 1.2], and

$$\widehat{\mathrm{vol}}(Y \ni y) = \inf_v \widehat{\mathrm{vol}}(v) = |G| \cdot \widehat{\mathrm{vol}}(X \ni x),$$

where v runs over the G-invariant valuations centered at y, and the second equality is [LX17, Theorem 2.7(1)]. For the F-volume, unlike for the normalized volume, a valuative viewpoint is missing [XZ26, Section 4.3], and our argument does not use an analogue of v_0. The goal of this paper is to answer Question 1.1 in the affirmative.

We have the following theorem.

Theorem 1.2. *Let $f : Y \to X$ be a finite surjective morphism between normal varieties of dimension $d \geq 1$, and let $x \in X$ and $y \in Y$ be closed points. Assume that f is étale in codimension one, X is strongly F-regular at x, and $f^{-1}(x) = \{y\}$. Then Y is strongly F-regular at y, and*

$$\mathrm{Fvol}(Y \ni y) = \deg(f) \cdot \mathrm{Fvol}(X \ni x) \quad \textit{and} \quad \mathrm{Fvol}_{\mathrm{HK}}(Y \ni y) = \deg(f) \cdot \mathrm{Fvol}_{\mathrm{HK}}(X \ni x).$$

The pair case also holds. For a $\mathbb{Q}$-divisor $B \geq 0$ on X such that $(X \ni x, B)$ is strongly F-regular (see Definition 2.2), the F-volume $\mathrm{Fvol}(X \ni x, B)$ and the Hilbert–Kunz F-volume $\mathrm{Fvol}_{\mathrm{HK}}(X \ni x, B)$ are defined by replacing $\mathrm{fpt}(X \ni x; \mathfrak{a})$ in the definitions above with the F-pure threshold $\mathrm{fpt}(X \ni x, B; \mathfrak{a})$ of $\mathfrak{a}$ with respect to $(X \ni x, B)$ (see Definition 2.4 and [XZ26, Section 4.3]).

Theorem 1.3. *Let $f : Y \to X$ be a finite surjective morphism between normal varieties of dimension $d \geq 1$, let $x \in X$ and $y \in Y$ be closed points, and let $B \geq 0$ be a $\mathbb{Q}$-divisor on X. Let $B_Y := f^*B$ be the divisorial pullback of B (see Definition 2.1). Assume that f is étale in codimension one, $(X \ni x, B)$ is strongly F-regular, and $f^{-1}(x) = \{y\}$. Then $(Y \ni y, B_Y)$ is strongly F-regular,*

$$\mathrm{Fvol}(Y \ni y, B_Y) = \deg(f) \cdot \mathrm{Fvol}(X \ni x, B),$$

and

$$\mathrm{Fvol}_{\mathrm{HK}}(Y \ni y, B_Y) = \deg(f) \cdot \mathrm{Fvol}_{\mathrm{HK}}(X \ni x, B).$$

We also prove the following theorem.

Theorem 1.4. *Let X be a normal variety of dimension $d \geq 1$, $x \in X$ a closed point, and $B \geq 0$ a $\mathbb{Q}$-divisor on X. Assume that $(X \ni x, B)$ is strongly F-regular. Then*

$$\mathrm{Fvol}_{\mathrm{HK}}(X \ni x, B) = \frac{\mathrm{Fvol}(X \ni x, B)}{d!}.$$

In particular, the formulas for $\mathrm{Fvol}_{\mathrm{HK}}$ in Theorems 1.2 and 1.3 follow from those for Fvol.

Sketch of the proof. By Theorem 1.4, it suffices to prove the formula for Fvol in Theorem 1.3. The proof has three steps.

(1) Completion preserves F-volumes (Proposition 6.2), and Construction-Proposition 6.3 yields a complete local domain, Galois over the completions of $\mathcal{O}_{X,x}$ and $\mathcal{O}_{Y,y}$ with groups G and H such that $|G| = \deg(f) \cdot |H|$. Thus the formula follows from Theorem 5.2 for both inclusions. Theorem 5.2 states

$$\mathrm{Fvol}(S, B_S) = |G| \cdot \mathrm{Fvol}(R, B)$$

for a Galois inclusion $R \subset S$ with group G (Set-up 5.1).

(2) By Theorem 4.1(1) and Proposition 5.5(3), based on Lemma 5.3, both sides equal the infimum of $e(\mathfrak{b})\,\mathrm{fpt}(S, B_S; \mathfrak{b})^d$ over the G-invariant $\mathfrak{m}_S$-primary ideals $\mathfrak{b}$.

(3) For an $\mathfrak{m}_S$-primary ideal $\mathfrak{a}$ and $g \in G$, the midpoint of the geodesic (Definition 3.1) between the filtration of powers of $\mathfrak{a}$, normalized to have F-pure threshold 1, and its translate by g has F-pure threshold at most 1 (Theorem 3.6), hence multiplicity at least $\mathrm{Fvol}(S, B_S)$ (Proposition 3.5(1)). Then Proposition 4.2 and Corollary 4.3 bound the multiplicity of the intersection of the translates by $u+(|G|-1)\sqrt{u^2 - \mathrm{Fvol}(S, B_S)^2}$, where $u = e(\mathfrak{a})\,\mathrm{fpt}(S, B_S; \mathfrak{a})^d$; as the intersection has F-pure threshold at most 1 (Definition-Proposition 3.4(2)), this proves Theorem 4.1.

We only obtain G-invariant ideals $\mathfrak{b}$ with $e(\mathfrak{b})\,\mathrm{fpt}(S, B_S; \mathfrak{b})^d$ close to $\mathrm{Fvol}(S, B_S)$, and never use an ideal attaining $\mathrm{Fvol}(S, B_S)$. Step (3) is the main difficulty: with $\ell_S(S/(\mathfrak{c}\cap\mathfrak{c}')) \le \ell_S(S/\mathfrak{c})+\ell_S(S/\mathfrak{c}')$ in place of Proposition 4.2, we would only obtain $\mathrm{Fvol}(R, B) \le \mathrm{Fvol}(S, B_S) \le |G| \cdot \mathrm{Fvol}(R, B)$.

(a) Multiplicities of intersections. The proof of the uniqueness of the valuation computing the normalized volume in [BLQ24, Corollary 1.3] ends with the equality case [BLQ24, Theorem 1.1(3)] of the concavity of the $(-1/d)$-th power of the multiplicity along geodesics, which does not apply to ideals $\mathfrak{a}$ with $e(\mathfrak{a})\,\mathrm{fpt}(S, B_S; \mathfrak{a})^d$ only close to $\mathrm{Fvol}(S, B_S)$. Proposition 4.2 plays the role of the equality case for such ideals. In its proof, for the geodesic $(\mathfrak{a}_{\bullet,t})_{t\in[0,1]}$ between two linearly bounded filtrations (Definition 3.1), the measure of [BLQ24, Section 4.4] writes the multiplicities of $\mathfrak{a}_{\bullet,0}$, $\mathfrak{a}_{\bullet,1}$, $\mathfrak{a}_{\bullet,1/2}$, and $\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}$ as the integrals of z^{-d}, $(1-z)^{-d}$, 2^d, and $\max\{z^{-d}, (1-z)^{-d}\}$ against one finite measure on $(0,1)$, and as $2^d \le (z(1-z))^{-d/2}$, two applications of the Cauchy–Schwarz inequality yield

$$e(\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}) \le u + \sqrt{u^2 - w^2}$$

if $e(\mathfrak{a}_{\bullet,0}) = e(\mathfrak{a}_{\bullet,1}) = u$ and $e(\mathfrak{a}_{\bullet,1/2}) = w$.

(b) Convexity of F-pure thresholds. The proof of [XZ20, Lemma 3.12], which yields the subadditivity of log canonical thresholds [XZ20, Theorem 3.11], uses the summation formula for multiplier ideals of a sum of several ideals, but the summation formula for big test ideals [Tak04, Theorem 3.1] has no boundary and concerns two ideals. Since one element computes all mixed big test ideals of (S, B_S) (Definition-Theorem 2.3(2)), the summation formula with a boundary and any number of ideals becomes an identity between ideals (Lemma 3.7(1)). As the F-pure threshold of a filtration is defined through its ideals (Definition-Proposition 3.4), we compare the geodesic with such sums at the level of ideals,

$$\mathfrak{a}_{m+1,t} \subset \sum_{i=0}^{m} \mathfrak{a}_{i/(1-t),0} \cap \mathfrak{a}_{(m-i)/t,1}$$

(**Step 1** of the proof of Theorem 3.6), and not through valuations as in [BLQ24, Proposition 5.1].

(c) Invariant ideals that are not extended. In step (2), Theorem 2.7 compares $\mathrm{fpt}(R, B; \mathfrak{a})$ with $\mathrm{fpt}(S, B_S; \mathfrak{a}S)$ for ideals $\mathfrak{a}$ of R, but a G-invariant ideal $\mathfrak{b}$ of S need not be extended from R, even after it is replaced by a power (Example 5.4). For $h \in \mathfrak{b}$, the coefficient of $T^{|G|-i}$ in $\prod_{g\in G}(T - g(h))$ belongs to $\mathfrak{b}^i \cap R$, so its (N/i)-th power belongs to $\mathfrak{b}^N \cap R$ for $N = \mathrm{lcm}(1, 2, \ldots, |G|)$, and hence $\mathfrak{b}^N$ is integral over $(\mathfrak{b}^N \cap R)S$ (Lemma 5.3). Integral dependence does not change multiplicities and F-pure thresholds, hence, by [HS06, Theorem 11.2.7] and Theorem 2.7,

$$|G| \cdot e(\mathfrak{b}^N \cap R)\,\mathrm{fpt}(R, B; \mathfrak{b}^N \cap R)^d = e(\mathfrak{b})\,\mathrm{fpt}(S, B_S; \mathfrak{b})^d$$

(Proposition 5.5(2)).

(d) Completion before Galois covers. The reduction to the Galois case in [XZ20, Section 4] passes to a Galois closure of $K(Y)/K(X)$, in which the integral closure of $\mathcal{O}_{X,x}$ is in general only semilocal, while Theorem 5.2 concerns local rings. We therefore complete first in step (1), as the integral closure of a complete local domain in a finite extension of the fraction field is local [HS06, Theorem 4.3.4].

The assumption that f is étale in codimension one is used through Theorem 2.7, and $f^{-1}(x) = \{y\}$ makes $\mathcal{O}_{X,x} \subset \mathcal{O}_{Y,y}$ module-finite of degree $\deg(f)$, which completion preserves (Construction-Proposition 6.3(1)).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce F-pure thresholds and F-volumes of pairs, and recall the behavior of big test ideals under finite morphisms that are étale in codimension one.

In Section 3 we define F-pure thresholds of filtrations and show that they are convex along geodesics (Theorem 3.6). In Section 4 we estimate the multiplicity of the intersection of two filtrations (Proposition 4.2), and prove that the infimum defining the F-volume of a strongly F-regular pair may be taken over the ideals invariant under any finite group of ring automorphisms preserving the boundary (Theorem 4.1(1)).

In Section 5 we prove the finite degree formula for F-volumes of pairs in the case of a Galois cover of complete local domains that is étale in codimension one (Theorem 5.2). In Section 6, for local rings of normal varieties at closed points, we show that big test ideals of pairs commute with completion, which implies that F-pure thresholds and F-volumes of pairs are preserved by completion (Proposition 6.2), and we construct a complete local Galois cover dominating the completion of a finite extension of such local rings that is étale in codimension one (Construction-Proposition 6.3). In Section 7 we prove Theorems 1.2, 1.3, and 1.4.

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2. Preliminaries

2.1. Divisors on normal domains.

Definition 2.1. Let R be a Noetherian normal domain. We write a $\mathbb{Q}$-divisor B on $\operatorname{Spec} R$ as $B = \sum_P b_P P$, where P runs over the height-one primes of R and $b_P \in \mathbb{Q}$. For a Weil divisor D on $\operatorname{Spec} R$, let

$$R(D) := \{z \in \operatorname{Frac} R \mid z = 0 \text{ or } \operatorname{div}(z) + D \geq 0\}.$$

For a ring automorphism g of R, let $g(B) := \sum_P b_P g(P)$.

Let $R \subset S$ be a module-finite inclusion of Noetherian normal domains, that is, S is a finitely generated R-module. For any height-one prime Q of S, the prime $Q \cap R$ has height one. Let ord_Q be the valuation of the discrete valuation ring S_Q, and let $t_{Q\cap R}$ be a uniformizer of $R_{Q\cap R}$. The *divisorial pullback* of B to $\operatorname{Spec} S$ is

$$B_S := \sum_Q b_{Q\cap R} \operatorname{ord}_Q(t_{Q\cap R}) Q,$$

where Q runs over the height-one primes of S. For a finite surjective morphism $f : Y \to X$ between normal varieties and a $\mathbb{Q}$-divisor $B = \sum_P b_P P$ on X, the *divisorial pullback* of B is

$$f^*B := \sum_Q b_{f(Q)} \operatorname{ord}_Q(t_{f(Q)}) Q,$$

where Q runs over the prime divisors of Y and $t_{f(Q)}$ is a uniformizer of $\mathcal{O}_{X,f(Q)}$. For a closed point $x \in X$, we denote by B_x the restriction of B to $\operatorname{Spec} \mathcal{O}_{X,x}$.

In Definition 2.1, the prime $Q \cap R$ has height one: since R is normal and S is a domain integral over R, incomparability implies that $Q \cap R$ is non-zero, and going-down implies that $Q \cap R$ has height at most one [HS06, Theorems 2.2.3 and 2.2.7]. For the notation f^*B, see [ST10, Section 2.2].

2.2. **Strongly F-regular triples and big test ideals.**

Definition 2.2 ([ST10, Definitions 6.2, 6.8, and 6.14])**.** Let R be a Noetherian ring of prime characteristic p. For an integer $e \geq 0$ and an R-module M, we denote by $F_*^e M$ the module M regarded as an R-module through the e-th iterate of the Frobenius morphism of R. We say that R is *F-finite* if $F_*^1 R$ is a finitely generated R-module.

Let $(R, \mathfrak{m}_R)$ be an F-finite normal local domain of characteristic p and $B \geq 0$ a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$. For an integer $e \geq 0$ and $q = p^e$, let $C_e^B \subset \operatorname{Hom}_R(F_*^e R, R)$ be the image of the restriction map

$$\operatorname{Hom}_R(F_*^e R(\lceil (q-1)B \rceil), R) \to \operatorname{Hom}_R(F_*^e R, R).$$

We call (R, B) a *pair*. For a non-zero ideal $\mathfrak{a} \subset R$ and a real number $t \geq 0$, we call $(R, B, \mathfrak{a}^t)$ a *triple*. We say that $(R, B, \mathfrak{a}^t)$ is *strongly F-regular* if for any non-zero $c \in R$, there exist an integer $e \geq 1$, a map $\varphi \in C_e^B$, and an element $u \in \mathfrak{a}^{\lceil t(q-1) \rceil}$, where $q = p^e$, such that $\varphi(F_*^e(cu)) = 1$. We say that (R, B) is *strongly F-regular* if (R, B, R^1) is strongly F-regular. We say that R is *strongly F-regular* if $(R, 0)$ is strongly F-regular.

Let X be a normal variety, $x \in X$ a closed point, and $B \geq 0$ a $\mathbb{Q}$-divisor on X. We say that $(X \ni x, B)$ is *strongly F-regular* if $(\mathcal{O}_{X,x}, B_x)$ is strongly F-regular. We say that X is *strongly F-regular at* x if $\mathcal{O}_{X,x}$ is strongly F-regular.

In Definition 2.2, the local ring $\mathcal{O}_{X,x}$ is F-finite, since k is perfect and $\mathcal{O}_{X,x}$ is essentially of finite type over k (see [ST10, Section 2.1]). The strong F-regularity of R in Definition 2.2 is that of [HH89, Section 3].

Definition-Theorem 2.3 ([ST21, Definition 2.10 and Remark 2.11] and [ST10, Definition 6.3 and Proposition 6.18(iv)])**.** Let $(R, \mathfrak{m}_R)$ be an F-finite normal local domain of characteristic p and $B \geq 0$ a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$.

(1) For any non-zero ideal $\mathfrak{a} \subset R$ and any real number $t \geq 0$, there exists a smallest non-zero ideal $\tau_b(R, B, \mathfrak{a}^t)$ of R such that

$$\varphi\big(F_*^e(\mathfrak{a}^{\lceil t(q-1) \rceil} \tau_b(R, B, \mathfrak{a}^t))\big) \subset \tau_b(R, B, \mathfrak{a}^t)$$

for any integer $e \geq 1$ and any $\varphi \in C_e^B$, where $q = p^e$. We call $\tau_b(R, B, \mathfrak{a}^t)$ the *big test ideal* of $(R, B, \mathfrak{a}^t)$. Moreover, $(R, B, \mathfrak{a}^t)$ is strongly F-regular if and only if $\tau_b(R, B, \mathfrak{a}^t) = R$.

(2) There exists a non-zero element $c \in R$ such that for any non-zero ideals $\mathfrak{a}_1, \dots, \mathfrak{a}_l \subset R$ and any real numbers $t_1, \dots, t_l \geq 0$, the *mixed big test ideal*

$$\tau_b(R, B, \mathfrak{a}_1^{t_1} \cdots \mathfrak{a}_l^{t_l}) := \sum_{e \geq 0} \sum_{\varphi \in C_e^B} \varphi\big(F_*^e(hc\mathfrak{a}_1^{\lceil t_1 q \rceil} \cdots \mathfrak{a}_l^{\lceil t_l q \rceil})\big), \quad q = p^e,$$

does not depend on the choice of a non-zero element $h \in R$. Moreover, $\tau_b(R, B, \mathfrak{a}_1^{t_1} \cdots \mathfrak{a}_l^{t_l})$ does not depend on the choice of such an element c. If $l = 1$, then $\tau_b(R, B, \mathfrak{a}_1^{t_1})$ is the big test ideal of (1).

Proof. (1) See [ST10, Definition 6.3 and Proposition 6.18(iv)].

(2) By [Sch08, Lemma 2.17], there exists a big sharp test element $c \in R$ for (R, B) in the sense of [BSTZ09, Definition 3.1]. For any non-zero $h \in R$, the element hc is again a big sharp test element for (R, B) by [BSTZ09, Definition 3.1]. Thus, by [ST21, Remark 2.11], the sum in (2) is equal to the test ideal of [ST21, Definition 2.10], which does not depend on h. If $c' \in R$ is another non-zero element satisfying (2), then the sum for c with $h = c'$ and the sum for c' with $h = c$ coincide, and hence the sums for c and for c' with $h = 1$ are equal. If $l = 1$ and $t_1 > 0$, then the test ideal of [ST21, Definition 2.10] is the big test ideal of [BSTZ09, Definition-Proposition 3.3], that is, the ideal generated by the big sharp test elements for $(R, B, \mathfrak{a}_1^{t_1})$. Since the term of index $p^e - 1$ of the family $\mathfrak{b}_\bullet := (\mathfrak{a}_1^{\lceil t_1 m \rceil})_{m \geq 0}$ is $\mathfrak{a}_1^{\lceil t_1 (p^e - 1) \rceil}$ for any integer $e \geq 0$, the big sharp test elements for $(R, B, \mathfrak{a}_1^{t_1})$ are the non-zero elements of the big test ideal J of [Sch08, Definitions 2.16 and 2.19] for B and $\mathfrak{b}_\bullet$, so the big test ideal of [BSTZ09, Definition-Proposition 3.3] is equal

to J. By [Sch08, Theorem 6.3], J is the smallest non-zero ideal satisfying the condition in (1), since the uniform compatibility condition of [Sch08, Theorem 6.3] for $\mathfrak{b}_\bullet$ is the condition in (1) for $\mathfrak{a} = \mathfrak{a}_1$ and $t = t_1$. If $t_1 = 0$, then replacing $(\mathfrak{a}_1, t_1)$ by $(R, 1)$ changes neither the sum in (2) nor the condition in (1), and hence the case $t_1 = 0$ follows from the case $t_1 > 0$. □

In Definition-Theorem 2.3(2), the ideal $\mathfrak{a}_i$ is raised to the power $\lceil t_i q \rceil$, while B is rounded as $\lceil (q-1)B \rceil$ through C_e^B. The point of Definition-Theorem 2.3(2) is that the formula holds with one element c, and with any non-zero multiple hc of c, for all ideals and all exponents simultaneously. Thus, without assuming that $K_R + B$ is $\mathbb{Q}$-Cartier, comparing mixed big test ideals of (R, B) reduces to containments between ideals.

2.3. **F-pure thresholds and F-volumes.**

Definition 2.4. Let $(R, \mathfrak{m}_R)$ be an F-finite normal local domain of characteristic p, and $B \geq 0$ a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$ such that (R, B) is strongly F-regular. For a non-zero ideal $\mathfrak{a} \subset R$, the *F-pure threshold* of $\mathfrak{a}$ with respect to (R, B) is

$$\operatorname{fpt}(R, B; \mathfrak{a}) := \sup\{t \geq 0 \mid (R, B, \mathfrak{a}^t) \text{ is strongly F-regular}\}.$$

If R is strongly F-regular, then we write $\operatorname{fpt}(R; \mathfrak{a}) := \operatorname{fpt}(R, 0; \mathfrak{a})$. Assume moreover that $\dim R = d \geq 1$. For an $\mathfrak{m}_R$-primary ideal $\mathfrak{a}$, we denote by

$$e_R(\mathfrak{a}) := \lim_{m \to +\infty} \frac{d! \ell_R(R/\mathfrak{a}^m)}{m^d}$$

the Hilbert–Samuel multiplicity of $\mathfrak{a}$, where ℓ_R is the length function on R-modules. For an $\mathfrak{m}_R$-primary ideal $\mathfrak{a}$, we call

$$e_{\mathrm{HK},R}(\mathfrak{a}) := \lim_{q \to +\infty} \frac{\ell_R(R/\mathfrak{a}^{[q]})}{q^d}$$

the *Hilbert–Kunz multiplicity* of $\mathfrak{a}$, where $q = p^e$ runs over the powers of p and $\mathfrak{a}^{[q]}$ is the ideal generated by the q-th powers of the elements of $\mathfrak{a}$. We drop the subscript R from $e_R(\mathfrak{a})$ and $e_{\mathrm{HK},R}(\mathfrak{a})$ when R is clear from the context. The *F-volume* of (R, B) is

$$\operatorname{Fvol}(R, B) := \inf_{\mathfrak{a}} e_R(\mathfrak{a}) \operatorname{fpt}(R, B; \mathfrak{a})^d,$$

where $\mathfrak{a}$ runs over the $\mathfrak{m}_R$-primary ideals of R. The *Hilbert–Kunz F-volume* of (R, B) is

$$\operatorname{Fvol}_{\mathrm{HK}}(R, B) := \inf_{\mathfrak{a}} e_{\mathrm{HK},R}(\mathfrak{a}) \operatorname{fpt}(R, B; \mathfrak{a})^d,$$

where $\mathfrak{a}$ runs over the $\mathfrak{m}_R$-primary ideals of R. If R is strongly F-regular, then we write $\operatorname{Fvol}(R) := \operatorname{Fvol}(R, 0)$ and $\operatorname{Fvol}_{\mathrm{HK}}(R) := \operatorname{Fvol}_{\mathrm{HK}}(R, 0)$.

Let X be a normal variety of dimension $d \geq 1$, $x \in X$ a closed point, and $B \geq 0$ a $\mathbb{Q}$-divisor on X such that $(X \ni x, B)$ is strongly F-regular. We write $\operatorname{fpt}(X \ni x, B; \mathfrak{a}) := \operatorname{fpt}(\mathcal{O}_{X,x}, B_x; \mathfrak{a})$, $\operatorname{Fvol}(X \ni x, B) := \operatorname{Fvol}(\mathcal{O}_{X,x}, B_x)$, and $\operatorname{Fvol}_{\mathrm{HK}}(X \ni x, B) := \operatorname{Fvol}_{\mathrm{HK}}(\mathcal{O}_{X,x}, B_x)$, and we drop B from the notation when $B = 0$.

Remark 2.5. In Definition 2.4, the limit defining $e_{\mathrm{HK},R}(\mathfrak{a})$ exists by [Mon83, Theorem 1.8] applied to $\widehat{R}$ (see also [Hun14, Theorem 3.12]). For $B = 0$, $\operatorname{fpt}(R; \mathfrak{a})$ is the F-pure threshold of $\mathfrak{a}$ in the sense of [TW03, Definition 2.1], by Lemma 2.6(2), [TW03, Definition 1.3(iii)], and [TW03, Proposition 2.2(5)]. For $\operatorname{Fvol}(X \ni x)$, see [LLX18, Definition 6.17]. For $\operatorname{Fvol}(X \ni x, B)$ and $\operatorname{Fvol}_{\mathrm{HK}}(X \ni x, B)$, see [XZ26, Section 4.3].

Lemma 2.6. *Let $(R, \mathfrak{m}_R)$ be an F-finite normal local domain of characteristic p, $B \geq 0$ a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$, and $\mathfrak{a} \subset \mathfrak{b}$ non-zero ideals of R. Assume that (R, B) is strongly F-regular. Then the following hold.*

(1) R is strongly F-regular.

(2) *For any real number $t \geq 0$, $(R, B, \mathfrak{a}^t)$ is strongly F-regular if and only if for any non-zero $c \in R$, there exist an integer $e \geq 1$, a map $\varphi \in C_e^B$, and an element $u \in \mathfrak{a}^{\lceil tq \rceil}$, where $q = p^e$, such that $\varphi(F_*^e(cu)) = 1$.*
(3) *$0 < \operatorname{fpt}(R, B; \mathfrak{a}) \leq \operatorname{fpt}(R, B; \mathfrak{b})$.*
(4) *$\operatorname{fpt}(R, B; \mathfrak{a}^m) = \operatorname{fpt}(R, B; \mathfrak{a})/m$ for any positive integer m.*
(5) *If $\mathfrak{a} \subset \mathfrak{m}_R$ and $\mathfrak{a}$ is generated by n elements, then $\operatorname{fpt}(R, B; \mathfrak{a}) \leq \operatorname{fpt}(R; \mathfrak{a}) \leq n$.*

Proof. (1) Since $C_e^B \subset \operatorname{Hom}_R(F_*^e R, R) = C_e^0$ for any integer $e \geq 1$, strong F-regularity of (R, B) implies strong F-regularity of $(R, 0)$, that is, of R.

(2) If the condition in (2) holds, then $(R, B, \mathfrak{a}^t)$ is strongly F-regular, since $\mathfrak{a}^{\lceil tq \rceil} \subset \mathfrak{a}^{\lceil t(q-1) \rceil}$. Conversely, assume that $(R, B, \mathfrak{a}^t)$ is strongly F-regular, and let $h \in \mathfrak{a}^{\lceil t \rceil}$ be a non-zero element. For any non-zero $c \in R$, the definition of strong F-regularity applied to ch yields an integer $e \geq 1$, a map $\varphi \in C_e^B$, and an element $u \in \mathfrak{a}^{\lceil t(q-1) \rceil}$, where $q = p^e$, such that $\varphi(F_*^e(chu)) = 1$. Since $\lceil t \rceil + \lceil t(q-1) \rceil \geq \lceil tq \rceil$, we have $hu \in \mathfrak{a}^{\lceil tq \rceil}$, which proves the condition in (2) (cf. [HY03, Remark 6.2(1)]).

(3) If $(R, B, \mathfrak{a}^t)$ is strongly F-regular for a real number $t \geq 0$, then so is $(R, B, \mathfrak{b}^t)$: for any non-zero $c \in R$, an integer e, a map φ, and an element u that satisfy the condition in Definition 2.2 for $(R, B, \mathfrak{a}^t)$ and c also satisfy it for $(R, B, \mathfrak{b}^t)$ and c, since $\mathfrak{a}^{\lceil t(q-1) \rceil} \subset \mathfrak{b}^{\lceil t(q-1) \rceil}$. Thus $\operatorname{fpt}(R, B; \mathfrak{a}) \leq \operatorname{fpt}(R, B; \mathfrak{b})$. For positivity, let $c \in R$ be a non-zero element satisfying Definition-Theorem 2.3(2), and let $h \in \mathfrak{a}$ be a non-zero element. Since (R, B) is strongly F-regular, there exist an integer $e \geq 1$, a map $\varphi \in C_e^B$, and an element $u \in R$ such that $\varphi(F_*^e(chu)) = 1$. As $hu \in \mathfrak{a} = \mathfrak{a}^{\lceil p^e/p^e \rceil}$, the formula of Definition-Theorem 2.3(2) for $l = 1$, $\mathfrak{a}_1 = \mathfrak{a}$, and $t_1 = 1/p^e$, with 1 as the multiplier of c, shows that $1 \in \tau_b(R, B, \mathfrak{a}^{1/p^e})$. Since $l = 1$, Definition-Theorem 2.3(1) implies that $(R, B, \mathfrak{a}^{1/p^e})$ is strongly F-regular, and hence $\operatorname{fpt}(R, B; \mathfrak{a}) \geq 1/p^e > 0$.

(4) Fix a non-zero element $h \in \mathfrak{a}^m$. For any real number $t \geq 0$ and any $q = p^e$ with $e \geq 1$, the inequalities

$$\lceil mt(q-1) \rceil \leq m \lceil t(q-1) \rceil \leq \lceil mt(q-1) \rceil + m$$

imply that

$$(\mathfrak{a}^m)^{\lceil t(q-1) \rceil} \subset \mathfrak{a}^{\lceil mt(q-1) \rceil} \quad \text{and} \quad h\mathfrak{a}^{\lceil mt(q-1) \rceil} \subset (\mathfrak{a}^m)^{\lceil t(q-1) \rceil}.$$

Thus, if $(R, B, (\mathfrak{a}^m)^t)$ is strongly F-regular, then so is $(R, B, \mathfrak{a}^{mt})$: for any non-zero $c \in R$, an integer e, a map φ, and an element u that satisfy the condition in Definition 2.2 for $(R, B, (\mathfrak{a}^m)^t)$ and c also satisfy it for $(R, B, \mathfrak{a}^{mt})$ and c. Conversely, if $(R, B, \mathfrak{a}^{mt})$ is strongly F-regular, then applying the definition to ch in place of c shows that $(R, B, (\mathfrak{a}^m)^t)$ is strongly F-regular. Taking suprema over t, we obtain (4). For $B = 0$, see also [TW03, Proposition 2.2(2)].

(5) Since $C_e^B \subset C_e^0$ for any integer $e \geq 1$, strong F-regularity of $(R, B, \mathfrak{a}^t)$ implies strong F-regularity of $(R, 0, \mathfrak{a}^t)$. Thus $\operatorname{fpt}(R, B; \mathfrak{a}) \leq \operatorname{fpt}(R; \mathfrak{a})$, where R is strongly F-regular by (1). Write $\mathfrak{a} = (a_1, \ldots, a_n)$, and let $h \in \mathfrak{a}^n$ be a non-zero element and $t \geq n$ a real number. For any $q = p^e$ with $e \geq 1$, we have

$$h\mathfrak{a}^{\lceil t(q-1) \rceil} \subset \mathfrak{a}^{nq} \subset \mathfrak{a}^{[q]} \subset \mathfrak{m}_R^{[q]},$$

where the first inclusion holds since $\lceil t(q-1) \rceil \geq n(q-1)$, and the second holds since any monomial of degree nq in $a_1, \ldots, a_n$ is divisible by a_i^q for some i. Any R-linear map $\varphi : F_*^e R \to R$ sends $F_*^e \mathfrak{m}_R^{[q]}$ into $\mathfrak{m}_R$, since $\varphi(F_*^e(z^q y)) = z\varphi(F_*^e y)$ for $z \in \mathfrak{m}_R$ and $y \in R$. Thus the condition in Definition 2.2 fails for $(R, 0, \mathfrak{a}^t)$ and $c = h$, that is, $(R, 0, \mathfrak{a}^t)$ is not strongly F-regular. Hence $\operatorname{fpt}(R; \mathfrak{a}) \leq n$. □

2.4. **Big test ideals under finite morphisms.**

Theorem 2.7 ([ST10, Corollaries 6.30, 6.31, and 7.7])**.** *Let $(R, \mathfrak{m}_R) \subset (S, \mathfrak{m}_S)$ be a module-finite inclusion of F-finite normal local domains of characteristic p, and $B \geq 0$ a $\mathbb{Q}$-divisor on* Spec R. *Assume that* Spec $S \to$ Spec R *is étale in codimension one, R is strongly F-regular, and either R*

and S are both essentially of finite type over k or R and S are both complete. Then the following hold.

(1) *For any non-zero ideal $\mathfrak{a} \subset R$ and any real number $t \geq 0$,*
$$\tau_b(S, B_S, (\mathfrak{a}S)^t) \cap R = \tau_b(R, B, \mathfrak{a}^t).$$
(2) *S is strongly F-regular, and for any non-zero ideal $\mathfrak{a} \subset R$ and any real number $t \geq 0$, $(R, B, \mathfrak{a}^t)$ is strongly F-regular if and only if $(S, B_S, (\mathfrak{a}S)^t)$ is strongly F-regular.*
(3) *If (R, B) is strongly F-regular, then (S, B_S) is strongly F-regular, and for any non-zero ideal $\mathfrak{a} \subset R$, we have* $\operatorname{fpt}(R, B; \mathfrak{a}) = \operatorname{fpt}(S, B_S; \mathfrak{a}S)$.

Proof. Since $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale in codimension one, [Gro71, Exp. I, Proposition 3.1] implies that the field extension $\operatorname{Frac} R \subset \operatorname{Frac} S$ is separable and that the ramification divisor in [ST10, Setting 6.27] is zero. Thus the boundary on $\operatorname{Spec} S$ in [ST10, Setting 6.27] is B_S, and $B_S \geq 0$ as $B \geq 0$. As R and S are both essentially of finite type over k or both complete, they have dualizing complexes [Har66, Chapter V, Section 10 and Corollary 2.3], and as they are local, they satisfy condition (!) of [ST10, Section 2.5]. By [ST10, Corollary 7.7], the trace map $\operatorname{Frac} S \to \operatorname{Frac} R$ sends S onto R. Thus (1) follows from [ST10, Corollary 6.30], and the equivalence in (2) follows from [ST10, Corollary 6.31] and Definition-Theorem 2.3(1). Taking $B = 0$, $\mathfrak{a} = R$, and $t = 1$ in the equivalence, we see that S is strongly F-regular. Taking $\mathfrak{a} = R$ and $t = 1$, we see that (S, B_S) is strongly F-regular if (R, B) is strongly F-regular. Since the equivalence holds for any $t \geq 0$, the sets whose suprema define $\operatorname{fpt}(R, B; \mathfrak{a})$ and $\operatorname{fpt}(S, B_S; \mathfrak{a}S)$ in Definition 2.4 are equal, which proves (3). □

Ideals of S that are not of the form $\mathfrak{a}S$ for an ideal $\mathfrak{a}$ of R are treated in Sections 4 and 5.

3. Filtrations and convexity of F-pure thresholds

In this section, we prove Theorem 3.6, the convexity of F-pure thresholds along geodesics between filtrations. Theorem 3.6 is an analogue in positive characteristic of the subadditivity of log canonical thresholds in [XZ20, Theorem 3.11], and it is used in Section 4 to bound the F-pure thresholds of midpoints of geodesics between translates of one filtration under a finite group of automorphisms of a local ring.

3.1. Filtrations and F-pure thresholds.

In this subsection, we define the F-pure threshold of a filtration and show that the F-volume can be computed by filtrations instead of ideals.

Definition 3.1 ([BLQ24, Definitions 2.1, 2.2, and 4.1])**.** Let $(R, \mathfrak{m}_R)$ be a Noetherian local domain with $\dim R = d \geq 1$. An *$\mathfrak{m}_R$-filtration* is a family $\mathfrak{a}_\bullet = (\mathfrak{a}_\lambda)_{\lambda \in \mathbb{R}_{>0}}$ of $\mathfrak{m}_R$-primary ideals of R such that

(1) $\mathfrak{a}_\lambda \subset \mathfrak{a}_\mu$ for any real numbers $\lambda > \mu > 0$,
(2) for any $\lambda > 0$, there exists $\epsilon_0 > 0$ such that $\mathfrak{a}_{\lambda-\epsilon} = \mathfrak{a}_\lambda$ for any $0 < \epsilon < \epsilon_0$, and
(3) $\mathfrak{a}_\lambda \mathfrak{a}_\mu \subset \mathfrak{a}_{\lambda+\mu}$ for any real numbers $\lambda, \mu > 0$.

We set $\mathfrak{a}_\lambda := R$ for $\lambda \leq 0$. An $\mathfrak{m}_R$-filtration $\mathfrak{a}_\bullet$ is *linearly bounded* if there exists a real number $b > 0$ such that $\mathfrak{a}_\lambda \subset \mathfrak{m}_R^{\lceil b\lambda \rceil}$ for any $\lambda > 0$. If the limit
$$e(\mathfrak{a}_\bullet) := \lim_{m \to +\infty} \frac{d! \ell_R(R/\mathfrak{a}_m)}{m^d}$$
over the positive integers m exists, then we call it the *multiplicity* of $\mathfrak{a}_\bullet$. For $\mathfrak{m}_R$-filtrations $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$, any $t \in (0,1)$, and any $\lambda > 0$, let
$$\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1} := (\mathfrak{a}_{\lambda,0} \cap \mathfrak{a}_{\lambda,1})_{\lambda>0} \quad \text{and} \quad \mathfrak{a}_{\lambda,t} := \sum_{(1-t)\mu + t\nu = \lambda} \mathfrak{a}_{\mu,0} \cap \mathfrak{a}_{\nu,1},$$
where μ and ν run over the real numbers. We call the family $(\mathfrak{a}_{\bullet,t})_{t\in[0,1]}$ the *geodesic* between $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$.

By [Cut15, Theorem 6.5], the multiplicity of any $\mathfrak{m}_R$-filtration exists if $\widehat{R}$ is reduced (see also [BLQ24, Section 2.4.2, (2.1)]). The midpoint $\mathfrak{a}_{\bullet,1/2}$ of a geodesic $(\mathfrak{a}_{\bullet,t})_{t\in[0,1]}$ and the intersection of the translates of a filtration under a finite group of automorphisms of R are in general not filtrations by powers of one ideal, and Proposition 3.5 shows that the F-volume does not change when ideals are replaced by filtrations. Linear boundedness guarantees that the multiplicity of an $\mathfrak{m}_R$-filtration $\mathfrak{a}_\bullet$ is positive, as needed for the measures in [BLQ24, Section 4] used in Section 4, and that the F-pure threshold of $\mathfrak{a}_\bullet$ is finite (Definition-Proposition 3.4), so that $e(\mathfrak{a}_\bullet)\operatorname{fpt}(R,B;\mathfrak{a}_\bullet)^d$ is a positive real number.

Lemma 3.2. *Let $(R,\mathfrak{m}_R)$ be a Noetherian local domain with $\dim R \geq 1$, let $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$ be linearly bounded $\mathfrak{m}_R$-filtrations, and let $t\in(0,1)$. Then $\mathfrak{a}_{\bullet,0}\cap\mathfrak{a}_{\bullet,1}$ and $\mathfrak{a}_{\bullet,t}$ are linearly bounded $\mathfrak{m}_R$-filtrations.*

Proof. Since $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$ are linearly bounded, and since $\mathfrak{m}_R^{\lceil b_1\lambda\rceil}\subset\mathfrak{m}_R^{\lceil b_2\lambda\rceil}$ for any real numbers $b_1\geq b_2>0$ and $\lambda>0$, there exists a real number $b>0$ such that $\mathfrak{a}_{\lambda,0}+\mathfrak{a}_{\lambda,1}\subset\mathfrak{m}_R^{\lceil b\lambda\rceil}$ for any $\lambda>0$.

The family $\mathfrak{a}_{\bullet,0}\cap\mathfrak{a}_{\bullet,1}$ satisfies Definition 3.1(1) and (3) because $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$ do, and it satisfies Definition 3.1(2) at any $\lambda>0$ with the minimum of the numbers ϵ_0 for $\mathfrak{a}_{\bullet,0}$ and for $\mathfrak{a}_{\bullet,1}$ at λ. Moreover, for any $\lambda>0$, $\mathfrak{a}_{\lambda,0}\cap\mathfrak{a}_{\lambda,1}$ is $\mathfrak{m}_R$-primary as an intersection of two $\mathfrak{m}_R$-primary ideals, and $\mathfrak{a}_{\lambda,0}\cap\mathfrak{a}_{\lambda,1}\subset\mathfrak{a}_{\lambda,0}\subset\mathfrak{m}_R^{\lceil b\lambda\rceil}$. Thus $\mathfrak{a}_{\bullet,0}\cap\mathfrak{a}_{\bullet,1}$ is a linearly bounded $\mathfrak{m}_R$-filtration.

Now we consider $\mathfrak{a}_{\bullet,t}$. Let $\lambda>0$, and let $\mu,\nu\in\mathbb{R}$ with $(1-t)\mu+t\nu=\lambda$. If $\mu<0$, then $\nu>\lambda/t$, and hence $\mathfrak{a}_{\mu,0}\cap\mathfrak{a}_{\nu,1}=\mathfrak{a}_{\nu,1}\subset\mathfrak{a}_{0,0}\cap\mathfrak{a}_{\lambda/t,1}$. If $\nu<0$, then $\mu>\lambda/(1-t)$, and hence $\mathfrak{a}_{\mu,0}\cap\mathfrak{a}_{\nu,1}=\mathfrak{a}_{\mu,0}\subset\mathfrak{a}_{\lambda/(1-t),0}\cap\mathfrak{a}_{0,1}$. Thus

$$\mathfrak{a}_{\lambda,t}=\sum_{\substack{\mu,\nu\geq0\\(1-t)\mu+t\nu=\lambda}}\mathfrak{a}_{\mu,0}\cap\mathfrak{a}_{\nu,1}\quad\text{for any }\lambda>0. \tag{3.1}$$

If $0<\lambda<\lambda'$, $\mu,\nu\geq0$, and $(1-t)\mu+t\nu=\lambda'$, then

$$\mathfrak{a}_{\mu,0}\cap\mathfrak{a}_{\nu,1}\subset\mathfrak{a}_{\lambda\mu/\lambda',0}\cap\mathfrak{a}_{\lambda\nu/\lambda',1}\subset\mathfrak{a}_{\lambda,t},$$

so (3.1) implies $\mathfrak{a}_{\lambda',t}\subset\mathfrak{a}_{\lambda,t}$. If $\lambda,\lambda'>0$, $\mu,\nu,\mu',\nu'\geq0$, $(1-t)\mu+t\nu=\lambda$, and $(1-t)\mu'+t\nu'=\lambda'$, then Definition 3.1(3) implies

$$(\mathfrak{a}_{\mu,0}\cap\mathfrak{a}_{\nu,1})(\mathfrak{a}_{\mu',0}\cap\mathfrak{a}_{\nu',1})\subset\mathfrak{a}_{\mu+\mu',0}\cap\mathfrak{a}_{\nu+\nu',1}\subset\mathfrak{a}_{\lambda+\lambda',t},$$

so (3.1) implies $\mathfrak{a}_{\lambda,t}\mathfrak{a}_{\lambda',t}\subset\mathfrak{a}_{\lambda+\lambda',t}$. If $\lambda>0$, $\mu,\nu\geq0$, and $(1-t)\mu+t\nu=\lambda$, then $\max(\mu,\nu)\geq\lambda$, and hence $\mathfrak{a}_{\mu,0}\cap\mathfrak{a}_{\nu,1}\subset\mathfrak{m}_R^{\lceil b\lambda\rceil}$. Thus by (3.1),

$$\mathfrak{a}_{\lambda,0}\cap\mathfrak{a}_{\lambda,1}\subset\mathfrak{a}_{\lambda,t}\subset\mathfrak{m}_R^{\lceil b\lambda\rceil}\subset\mathfrak{m}_R\quad\text{for any }\lambda>0,$$

and hence $\mathfrak{a}_{\lambda,t}$ is $\mathfrak{m}_R$-primary.

It remains to show that $\mathfrak{a}_{\bullet,t}$ satisfies Definition 3.1(2). Fix $\lambda_0>0$, let $M:=\lambda_0/\min(t,1-t)$, and let $N\geq1$ be an integer such that $\mathfrak{m}_R^N\subset\mathfrak{a}_{M,0}\cap\mathfrak{a}_{M,1}$. Let $i\in\{0,1\}$. The ideals $\mathfrak{a}_{\mu,i}$ with $\mu\in[0,M]$ form a chain of ideals containing $\mathfrak{m}_R^N$, so the set $\{\mathfrak{a}_{\mu,i}\mid\mu\in[0,M]\}$ has at most $\ell_R(R/\mathfrak{m}_R^N)+1$ elements. By Definition 3.1(1) and (2), for any $\mu\in[0,M]$, the set $\{\mu'\in[0,M]\mid\mathfrak{a}_{\mu',i}=\mathfrak{a}_{\mu,i}\}$ has a maximum. Thus

$$\Lambda_i:=\left\{\max\{\mu'\in[0,M]\mid\mathfrak{a}_{\mu',i}=\mathfrak{a}_{\mu,i}\}\;\middle|\;\mu\in[0,M]\right\}$$

is a finite set.

Let $0<\lambda\leq\lambda_0$. If $\mu,\nu\geq0$ and $(1-t)\mu+t\nu=\lambda$, then $\mu,\nu\leq M$, so there exist $\mu'\in\Lambda_0$ and $\nu'\in\Lambda_1$ such that $\mu'\geq\mu$, $\nu'\geq\nu$, $\mathfrak{a}_{\mu',0}=\mathfrak{a}_{\mu,0}$, and $\mathfrak{a}_{\nu',1}=\mathfrak{a}_{\nu,1}$, and then $(1-t)\mu'+t\nu'\geq\lambda$. Conversely, if $\mu\in\Lambda_0$, $\nu\in\Lambda_1$, and $(1-t)\mu+t\nu\geq\lambda$, then $s:=\lambda/((1-t)\mu+t\nu)\in(0,1]$ satisfies

$(1-t)s\mu+ts\nu=\lambda$, and hence $\mathfrak{a}_{\mu,0}\cap\mathfrak{a}_{\nu,1}\subset\mathfrak{a}_{s\mu,0}\cap\mathfrak{a}_{s\nu,1}\subset\mathfrak{a}_{\lambda,t}$. Thus by (3.1),

$$\mathfrak{a}_{\lambda,t}=\sum_{\substack{\mu\in\Lambda_0,\nu\in\Lambda_1\\(1-t)\mu+t\nu\geq\lambda}}\mathfrak{a}_{\mu,0}\cap\mathfrak{a}_{\nu,1}\quad\text{for any } 0<\lambda\leq\lambda_0.$$

Since Λ_0 and Λ_1 are finite, there exists $0<\epsilon_0<\lambda_0$ such that the interval $(\lambda_0-\epsilon_0,\lambda_0)$ contains no number $(1-t)\mu+t\nu$ with $\mu\in\Lambda_0$ and $\nu\in\Lambda_1$. Thus for any $0<\epsilon<\epsilon_0$ and any $(\mu,\nu)\in\Lambda_0\times\Lambda_1$, the inequalities $(1-t)\mu+t\nu\geq\lambda_0-\epsilon$ and $(1-t)\mu+t\nu\geq\lambda_0$ are equivalent, so $\mathfrak{a}_{\lambda_0-\epsilon,t}=\mathfrak{a}_{\lambda_0,t}$. Therefore $\mathfrak{a}_{\bullet,t}$ is a linearly bounded $\mathfrak{m}_R$-filtration. □

Set-up 3.3. Let R, $\mathfrak{m}_R$, d, and B be as follows.

(1) $(R,\mathfrak{m}_R)$ is an excellent F-finite normal local domain of characteristic p with $\dim R=d\geq 1$.
(2) $B\geq 0$ is a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$ such that (R,B) is strongly F-regular.

We refer to [Gro65, (7.8.2)] for the definition of excellent rings. Local rings of varieties at closed points and complete local rings are excellent [Gro65, (7.8.3)(ii) and (iii)]. In Set-up 3.3, $\widehat{R}$ is a normal domain [Gro65, (7.8.3)(v)]. Thus $\widehat{R}$ is reduced, as assumed in [Cut15, Theorem 6.5], and $\widehat{R}$ is a domain, as assumed in [BLQ24, Section 4].

Definition-Proposition 3.4. Notation and conditions as in Set-up 3.3. Let $\mathfrak{a}_\bullet$ be a linearly bounded $\mathfrak{m}_R$-filtration. Then the limits and suprema

$$\lim_{m\to+\infty}m\operatorname{fpt}(R,B;\mathfrak{a}_m),\quad\sup_{m\geq1}m\operatorname{fpt}(R,B;\mathfrak{a}_m),\quad\sup_{\lambda>0}\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda),\quad\lim_{\lambda\to+\infty}\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda),$$

where m runs over the positive integers and λ over the positive real numbers, exist and are equal to a positive real number, which we denote by $\operatorname{fpt}(R,B;\mathfrak{a}_\bullet)$ and call the *F-pure threshold* of $\mathfrak{a}_\bullet$ with respect to (R,B). Moreover, the following hold.

(1) The multiplicity $e(\mathfrak{a}_\bullet)$ exists, $0<e(\mathfrak{a}_\bullet)<+\infty$, and
$$e(\mathfrak{a}_\bullet)=\lim_{m\to+\infty}\frac{e_R(\mathfrak{a}_m)}{m^d}.$$
(2) If $\mathfrak{b}_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration such that $\mathfrak{b}_\lambda\subset\mathfrak{a}_\lambda$ for any $\lambda>0$, then $\operatorname{fpt}(R,B;\mathfrak{b}_\bullet)\leq\operatorname{fpt}(R,B;\mathfrak{a}_\bullet)$.

Proof. Let $b>0$ be a real number such that $\mathfrak{a}_\lambda\subset\mathfrak{m}_R^{\lceil b\lambda\rceil}$ for any $\lambda>0$. By Lemma 2.6(5), $\operatorname{fpt}(R,B;\mathfrak{m}_R)<+\infty$. Thus by Lemma 2.6(3) and (4), for any $\lambda>0$,

$$\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda)\leq\lambda\operatorname{fpt}(R,B;\mathfrak{m}_R^{\lceil b\lambda\rceil})=\frac{\lambda}{\lceil b\lambda\rceil}\operatorname{fpt}(R,B;\mathfrak{m}_R)\leq\frac{\operatorname{fpt}(R,B;\mathfrak{m}_R)}{b}<+\infty.\tag{3.2}$$

Fix a real number $\lambda>0$. For any real number $\mu>0$, let $j:=\lceil\mu/\lambda\rceil$. Since $j\lambda\geq\mu$, Definition 3.1(1) and (3) imply $\mathfrak{a}_\lambda^j\subset\mathfrak{a}_{j\lambda}\subset\mathfrak{a}_\mu$. Thus by Lemma 2.6(3) and (4),

$$\mu\operatorname{fpt}(R,B;\mathfrak{a}_\mu)\geq\mu\operatorname{fpt}(R,B;\mathfrak{a}_\lambda^j)=\frac{\mu}{\lceil\mu/\lambda\rceil}\operatorname{fpt}(R,B;\mathfrak{a}_\lambda).$$

Since $\mu/\lambda\leq\lceil\mu/\lambda\rceil<\mu/\lambda+1$, the right-hand side tends to $\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda)$ as $\mu\to+\infty$. Since $\lambda>0$ is arbitrary,

$$\sup_{\lambda>0}\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda)\leq\liminf_{\mu\to+\infty}\mu\operatorname{fpt}(R,B;\mathfrak{a}_\mu)\leq\limsup_{\mu\to+\infty}\mu\operatorname{fpt}(R,B;\mathfrak{a}_\mu)\leq\sup_{\lambda>0}\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda),$$

where μ runs over the positive real numbers. Hence $\lim_{\mu\to+\infty}\mu\operatorname{fpt}(R,B;\mathfrak{a}_\mu)$ exists and is equal to $\sup_{\lambda>0}\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda)$, which is finite by (3.2) and at least $\operatorname{fpt}(R,B;\mathfrak{a}_1)>0$ by Lemma 2.6(3). Restricting μ to the positive integers m, the limit $\lim_{m\to+\infty}m\operatorname{fpt}(R,B;\mathfrak{a}_m)$ exists and is equal to $\sup_{\lambda>0}\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda)$, and

$$\lim_{m\to+\infty}m\operatorname{fpt}(R,B;\mathfrak{a}_m)\leq\sup_{m\geq1}m\operatorname{fpt}(R,B;\mathfrak{a}_m)\leq\sup_{\lambda>0}\lambda\operatorname{fpt}(R,B;\mathfrak{a}_\lambda).$$

Thus all the limits and suprema in the statement exist and are equal to a positive real number.

(1) Since R is an excellent normal local domain, $\widehat{R}$ is a normal domain [Gro65, (7.8.3)(v)], and in particular $\widehat{R}$ is reduced. Moreover, $\mathfrak{a}_0 = R$, $\mathfrak{a}_m\mathfrak{a}_{m'} \subset \mathfrak{a}_{m+m'}$ for any integers $m, m' \geq 0$, and $\mathfrak{a}_m$ is $\mathfrak{m}_R$-primary for any integer $m \geq 1$. Thus by [Cut15, Theorem 6.5], $e(\mathfrak{a}_\bullet)$ exists and is equal to $\lim_{m\to+\infty} e_R(\mathfrak{a}_m)/m^d$. Since $\mathfrak{a}_1^m \subset \mathfrak{a}_m \subset \mathfrak{m}_R^{\lceil bm\rceil}$ for any integer $m \geq 1$, and since $\lceil bm\rceil/m \to b$ as $m \to +\infty$, Definition 2.4 implies

$$b^d e_R(\mathfrak{m}_R) = \lim_{m\to+\infty} \frac{d!\ell_R(R/\mathfrak{m}_R^{\lceil bm\rceil})}{m^d} \leq e(\mathfrak{a}_\bullet) \leq \lim_{m\to+\infty} \frac{d!\ell_R(R/\mathfrak{a}_1^m)}{m^d} = e_R(\mathfrak{a}_1) < +\infty.$$

Moreover, $e_R(\mathfrak{m}_R) > 0$, since $\ell_R(R/\mathfrak{m}_R^n)$ agrees with a polynomial in n of degree d for any sufficiently large integer n [HS06, Theorem 11.1.3]. Thus $0 < e(\mathfrak{a}_\bullet) < +\infty$.

(2) By Lemma 2.6(3), $\lambda \operatorname{fpt}(R, B; \mathfrak{b}_\lambda) \leq \lambda \operatorname{fpt}(R, B; \mathfrak{a}_\lambda)$ for any $\lambda > 0$. Taking the supremum over $\lambda > 0$ implies (2). □

Proposition 3.5. *Notation and conditions as in Set-up 3.3. Then the following hold.*

(1) For any linearly bounded $\mathfrak{m}_R$-filtration $\mathfrak{a}_\bullet$, we have $\operatorname{Fvol}(R, B) \leq e(\mathfrak{a}_\bullet) \operatorname{fpt}(R, B; \mathfrak{a}_\bullet)^d$.

(2) Let $\mathfrak{a}$ be an $\mathfrak{m}_R$-primary ideal and $s := \operatorname{fpt}(R, B; \mathfrak{a})$. Then $\mathfrak{b}_\lambda := \mathfrak{a}^{\lceil s\lambda\rceil}$ for $\lambda > 0$ defines a linearly bounded $\mathfrak{m}_R$-filtration $\mathfrak{b}_\bullet$ such that $\operatorname{fpt}(R, B; \mathfrak{b}_\bullet) = 1$ and $e(\mathfrak{b}_\bullet) = e_R(\mathfrak{a})s^d$.

(3) We have

$$\operatorname{Fvol}(R, B) = \inf_{\mathfrak{a}_\bullet} e(\mathfrak{a}_\bullet) \operatorname{fpt}(R, B; \mathfrak{a}_\bullet)^d = \inf_{\operatorname{fpt}(R,B;\mathfrak{a}_\bullet)=1} e(\mathfrak{a}_\bullet),$$

where $\mathfrak{a}_\bullet$ runs over the linearly bounded $\mathfrak{m}_R$-filtrations.

Proof. (1) For any integer $m \geq 1$, $\mathfrak{a}_m$ is $\mathfrak{m}_R$-primary, so Definition 2.4 implies

$$\operatorname{Fvol}(R, B) \leq e_R(\mathfrak{a}_m) \operatorname{fpt}(R, B; \mathfrak{a}_m)^d = \frac{e_R(\mathfrak{a}_m)}{m^d} \big(m \operatorname{fpt}(R, B; \mathfrak{a}_m)\big)^d.$$

By Definition-Proposition 3.4, the right-hand side tends to $e(\mathfrak{a}_\bullet) \operatorname{fpt}(R, B; \mathfrak{a}_\bullet)^d$ as $m \to +\infty$. Thus (1) holds.

(2) By Lemma 2.6(3) and (5), $0 < s < +\infty$. For any real numbers $\lambda, \mu > 0$, we have $\lceil s\lambda\rceil + \lceil s\mu\rceil \geq \lceil s(\lambda+\mu)\rceil$, and $\lceil s\lambda\rceil \geq \lceil s\mu\rceil$ if $\lambda > \mu$. For any $\lambda > 0$, we have $\lceil s(\lambda-\epsilon)\rceil = \lceil s\lambda\rceil$ for any $0 < \epsilon < (s\lambda - \lceil s\lambda\rceil + 1)/s$. Thus $\mathfrak{b}_\bullet$ satisfies Definition 3.1(1)–(3). Since $\lceil s\lambda\rceil \geq 1$ and $\mathfrak{a} \subset \mathfrak{m}_R$, $\mathfrak{b}_\lambda$ is $\mathfrak{m}_R$-primary and $\mathfrak{b}_\lambda \subset \mathfrak{m}_R^{\lceil s\lambda\rceil}$ for any $\lambda > 0$. Thus $\mathfrak{b}_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration. By Definition-Proposition 3.4 and Lemma 2.6(4),

$$\operatorname{fpt}(R, B; \mathfrak{b}_\bullet) = \lim_{m\to+\infty} m \operatorname{fpt}(R, B; \mathfrak{a}^{\lceil sm\rceil}) = \lim_{m\to+\infty} \frac{ms}{\lceil sm\rceil} = 1.$$

By [HS06, Proposition 11.2.9(1)], $e_R(\mathfrak{a}^j) = j^d e_R(\mathfrak{a})$ for any positive integer j. Thus by Definition-Proposition 3.4(1),

$$e(\mathfrak{b}_\bullet) = \lim_{m\to+\infty} \frac{\lceil sm\rceil^d e_R(\mathfrak{a})}{m^d} = e_R(\mathfrak{a})s^d.$$

(3) By (1),

$$\operatorname{Fvol}(R, B) \leq \inf_{\mathfrak{a}_\bullet} e(\mathfrak{a}_\bullet) \operatorname{fpt}(R, B; \mathfrak{a}_\bullet)^d \leq \inf_{\operatorname{fpt}(R,B;\mathfrak{a}_\bullet)=1} e(\mathfrak{a}_\bullet),$$

where the second inequality holds because $e(\mathfrak{a}_\bullet) \operatorname{fpt}(R, B; \mathfrak{a}_\bullet)^d = e(\mathfrak{a}_\bullet)$ if $\operatorname{fpt}(R, B; \mathfrak{a}_\bullet) = 1$. By (2), for any $\mathfrak{m}_R$-primary ideal $\mathfrak{a}$,

$$\inf_{\operatorname{fpt}(R,B;\mathfrak{a}_\bullet)=1} e(\mathfrak{a}_\bullet) \leq e_R(\mathfrak{a}) \operatorname{fpt}(R, B; \mathfrak{a})^d.$$

Taking the infimum over $\mathfrak{a}$, Definition 2.4 implies

$$\inf_{\operatorname{fpt}(R,B;\mathfrak{a}_\bullet)=1} e(\mathfrak{a}_\bullet) \leq \operatorname{Fvol}(R, B).$$

Thus (3) holds. □

3.2. **Convexity along geodesics.** In this subsection, we prove Theorem 3.6.

Theorem 3.6. *Notation and conditions as in Set-up 3.3. Let $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$ be linearly bounded $\mathfrak{m}_R$-filtrations, and let $t \in (0,1)$. Then*

$$\operatorname{fpt}(R,B;\mathfrak{a}_{\bullet,t}) \leq (1-t)\operatorname{fpt}(R,B;\mathfrak{a}_{\bullet,0}) + t\operatorname{fpt}(R,B;\mathfrak{a}_{\bullet,1}).$$

Since the F-pure threshold of a filtration is defined in terms of the F-pure thresholds of the ideals of the filtration (Definition-Proposition 3.4), **Step 1** of the proof compares the ideals of $\mathfrak{a}_{\bullet,t}$ with finite sums of intersections of ideals of $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$.

The proof of Theorem 3.6 expands the big test ideal of a sum of finitely many ideals with respect to (R,B). When $B=0$ and there are two ideals, Lemma 3.7(1) is [Tak04, Theorem 3.1] for F-finite normal local domains.

Lemma 3.7. *Let $(R,\mathfrak{m}_R)$ be an F-finite normal local domain of characteristic p, $B \geq 0$ a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$, and $\mathfrak{a}_1,\dots,\mathfrak{a}_l$ non-zero ideals of R. Then the following hold.*

(1) For any real number $u > 0$,

$$\tau_b(R,B,(\mathfrak{a}_1+\cdots+\mathfrak{a}_l)^u) = \sum_{(u_1,\dots,u_l)} \tau_b(R,B,\mathfrak{a}_1^{u_1}\cdots\mathfrak{a}_l^{u_l}),$$

where $(u_1,\dots,u_l)$ runs over the tuples of real numbers $u_i \geq 0$ with $u_1+\cdots+u_l = u$.

(2) Let $\mathfrak{b}$ be a non-zero ideal of R and $m_1,\dots,m_l$ positive integers such that $\mathfrak{a}_i^{m_i} \subset \mathfrak{b}$ for any i. Then for any real numbers $u_1,\dots,u_l \geq 0$,

$$\tau_b(R,B,\mathfrak{a}_1^{u_1}\cdots\mathfrak{a}_l^{u_l}) \subset \tau_b(R,B,\mathfrak{b}^v), \quad v := \sum_{i=1}^{l} \frac{u_i}{m_i}.$$

Proof. Let $c \in R$ be a non-zero element satisfying Definition-Theorem 2.3(2). By Definition-Theorem 2.3(2), any mixed big test ideal below is computed by the formula there with c and with any non-zero $h \in R$, and for one ideal the formula computes the big test ideal of Definition-Theorem 2.3(1). Let $e \geq 0$ be an integer, $q := p^e$, and $\varphi \in C_e^B$.

(1) We have

$$(\mathfrak{a}_1+\cdots+\mathfrak{a}_l)^{\lceil uq \rceil} = \sum_{(k_1,\dots,k_l)} \mathfrak{a}_1^{k_1}\cdots\mathfrak{a}_l^{k_l},$$

where $(k_1,\dots,k_l)$ runs over the tuples of non-negative integers with $k_1+\cdots+k_l = \lceil uq \rceil$. Fix such a tuple. Since $u > 0$, $\lceil uq \rceil \geq 1$, so there exists $1 \leq j \leq l$ such that $k_j \geq 1$. Let $u_j := (k_j + uq - \lceil uq \rceil)/q$, and let $u_i := k_i/q$ for $i \neq j$. Since $0 \leq \lceil uq \rceil - uq < 1$, we have $u_i \geq 0$ and $\lceil u_i q \rceil = k_i$ for any i, and $u_1+\cdots+u_l = u$. Thus by Definition-Theorem 2.3(2) with $h=1$,

$$\varphi\big(F_*^e(c\mathfrak{a}_1^{k_1}\cdots\mathfrak{a}_l^{k_l})\big) \subset \tau_b(R,B,\mathfrak{a}_1^{u_1}\cdots\mathfrak{a}_l^{u_l}).$$

Summing over the tuples $(k_1,\dots,k_l)$, the maps $\varphi \in C_e^B$, and the integers $e \geq 0$, Definition-Theorem 2.3(2) with $h=1$ implies that the left-hand side of (1) is contained in the right-hand side of (1).

Conversely, let $u_1,\dots,u_l \geq 0$ be real numbers with $u_1+\cdots+u_l = u$. Then

$$\mathfrak{a}_1^{\lceil u_1 q \rceil}\cdots\mathfrak{a}_l^{\lceil u_l q \rceil} \subset (\mathfrak{a}_1+\cdots+\mathfrak{a}_l)^{\lceil uq \rceil},$$

since $\lceil u_1 q \rceil + \cdots + \lceil u_l q \rceil \geq \lceil uq \rceil$, and hence

$$\varphi\big(F_*^e(c\mathfrak{a}_1^{\lceil u_1 q \rceil}\cdots\mathfrak{a}_l^{\lceil u_l q \rceil})\big) \subset \varphi\big(F_*^e(c(\mathfrak{a}_1+\cdots+\mathfrak{a}_l)^{\lceil uq \rceil})\big).$$

Summing over $\varphi \in C_e^B$ and $e \geq 0$, Definition-Theorem 2.3(2) with $h=1$ implies

$$\tau_b(R,B,\mathfrak{a}_1^{u_1}\cdots\mathfrak{a}_l^{u_l}) \subset \tau_b(R,B,(\mathfrak{a}_1+\cdots+\mathfrak{a}_l)^u).$$

Thus (1) holds.

(2) For any $1 \le i \le l$, since $\mathfrak{a}_i^{m_i} \subset \mathfrak{b}$,

$$\mathfrak{a}_i^{\lceil u_i q \rceil} \subset \left(\mathfrak{a}_i^{m_i}\right)^{\lfloor \lceil u_i q \rceil / m_i \rfloor} \subset \mathfrak{b}^{\lfloor \lceil u_i q \rceil / m_i \rfloor} \quad \text{and} \quad \left\lfloor \frac{\lceil u_i q \rceil}{m_i} \right\rfloor \ge \frac{u_i q}{m_i} - 1.$$

Thus $\mathfrak{a}_1^{\lceil u_1 q \rceil} \cdots \mathfrak{a}_l^{\lceil u_l q \rceil} \subset \mathfrak{b}^N$, where $N := \sum_{i=1}^l \lfloor \lceil u_i q \rceil / m_i \rfloor \ge vq - l$. Let h be a non-zero element of $\mathfrak{b}^{l+1}$. Since $N + l + 1 \ge vq + 1 > \lceil vq \rceil$,

$$h\mathfrak{a}_1^{\lceil u_1 q \rceil} \cdots \mathfrak{a}_l^{\lceil u_l q \rceil} \subset \mathfrak{b}^{N+l+1} \subset \mathfrak{b}^{\lceil vq \rceil},$$

and hence

$$\varphi\left(F_*^e(hc\mathfrak{a}_1^{\lceil u_1 q \rceil} \cdots \mathfrak{a}_l^{\lceil u_l q \rceil})\right) \subset \varphi\left(F_*^e(c\mathfrak{b}^{\lceil vq \rceil})\right).$$

Summing over $\varphi \in C_e^B$ and $e \ge 0$, Definition-Theorem 2.3(2), with h for the left-hand side and with $h = 1$ for the right-hand side, implies (2). □

Proof of Theorem 3.6. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that $\mathfrak{a}_{m+1,t} \subset \mathfrak{d}_m$ for the integer m and the ideal $\mathfrak{d}_m$ defined below.

Let $m \ge 1$ be an integer, which is fixed in **Steps 1 and 2**. For any integer $0 \le i \le m$, let

$$\mathfrak{c}_i := \mathfrak{a}_{i/(1-t),0} \cap \mathfrak{a}_{(m-i)/t,1},$$

and let $\mathfrak{d}_m := \mathfrak{c}_0 + \cdots + \mathfrak{c}_m$. Since $\mathfrak{a}_{0,0} = \mathfrak{a}_{0,1} = R$, we have $\mathfrak{c}_0 = \mathfrak{a}_{m/t,1}$ and $\mathfrak{c}_m = \mathfrak{a}_{m/(1-t),0}$. Let $\mu, \nu \in \mathbb{R}$ with $(1-t)\mu + t\nu = m+1$. If $\mu < 0$, then $t\nu > m+1$, and hence $\mathfrak{a}_{\mu,0} \cap \mathfrak{a}_{\nu,1} = \mathfrak{a}_{\nu,1}$ is contained in $\mathfrak{a}_{m/t,1} = \mathfrak{c}_0$. If $\nu < 0$, then $(1-t)\mu > m+1$, and hence $\mathfrak{a}_{\mu,0} \cap \mathfrak{a}_{\nu,1} = \mathfrak{a}_{\mu,0}$ is contained in $\mathfrak{a}_{m/(1-t),0} = \mathfrak{c}_m$. Suppose that $\mu, \nu \ge 0$, and let $i := \min(\lfloor (1-t)\mu \rfloor, m)$. Then $0 \le i \le (1-t)\mu$. If $i = \lfloor (1-t)\mu \rfloor$, then $m - i < m + 1 - (1-t)\mu = t\nu$, and if $i = m$, then $m - i = 0 \le t\nu$. Thus $(m-i)/t \le \nu$, and Definition 3.1(1) implies

$$\mathfrak{a}_{\mu,0} \cap \mathfrak{a}_{\nu,1} \subset \mathfrak{a}_{i/(1-t),0} \cap \mathfrak{a}_{(m-i)/t,1} = \mathfrak{c}_i.$$

Since $\mathfrak{a}_{m+1,t}$ is the sum of the ideals $\mathfrak{a}_{\mu,0} \cap \mathfrak{a}_{\nu,1}$ with $(1-t)\mu + t\nu = m+1$, we obtain $\mathfrak{a}_{m+1,t} \subset \mathfrak{d}_m$.

Step 2. In this step, we show that $m \operatorname{fpt}(R, B; \mathfrak{d}_m) \le \alpha + \beta$, where

$$\alpha := (1-t) \operatorname{fpt}(R, B; \mathfrak{a}_{\bullet,0}) \quad \text{and} \quad \beta := t \operatorname{fpt}(R, B; \mathfrak{a}_{\bullet,1}).$$

The ideals $\mathfrak{c}_0, \ldots, \mathfrak{c}_m$ are $\mathfrak{m}_R$-primary, and hence so is $\mathfrak{d}_m$. Let $c \in R$ be a non-zero element satisfying Definition-Theorem 2.3(2). Let $u > (\alpha + \beta)/m$ be a real number. By Lemma 3.7(1),

$$\tau_b(R, B, \mathfrak{d}_m^u) = \sum_{(u_0, \ldots, u_m)} \tau_b(R, B, \mathfrak{c}_0^{u_0} \cdots \mathfrak{c}_m^{u_m}),$$

where $(u_0, \ldots, u_m)$ runs over the tuples of real numbers $u_i \ge 0$ with $u_0 + \cdots + u_m = u$. Fix such a tuple, and let

$$x := \sum_{i=0}^m i u_i \quad \text{and} \quad y := \sum_{i=0}^m (m-i) u_i.$$

Then $x + y = mu > \alpha + \beta$, so $x > \alpha$ or $y > \beta$.

Suppose that $x > \alpha$. For any $1 \le i \le m$, Definition 3.1(3) implies

$$\mathfrak{c}_i^{m!/i} \subset \mathfrak{a}_{i/(1-t),0}^{m!/i} \subset \mathfrak{a}_{m!/(1-t),0},$$

and $\sum_{i=1}^m i u_i / m! = x/m!$. Since $\mathfrak{c}_0^{\lceil u_0 q \rceil} \subset R$ for any $q = p^e$, Definition-Theorem 2.3(2) with c and $h = 1$ on both sides, and Lemma 3.7(2) applied to the ideals $\mathfrak{c}_1, \ldots, \mathfrak{c}_m$, the ideal $\mathfrak{a}_{m!/(1-t),0}$, and the integers $m!/1, \ldots, m!/m$, imply

$$\tau_b(R, B, \mathfrak{c}_0^{u_0} \cdots \mathfrak{c}_m^{u_m}) \subset \tau_b(R, B, \mathfrak{c}_1^{u_1} \cdots \mathfrak{c}_m^{u_m}) \subset \tau_b(R, B, \mathfrak{a}_{m!/(1-t),0}^{x/m!}).$$

By the supremum formula in Definition-Proposition 3.4 with $\lambda = m!/(1-t)$,

$$\operatorname{fpt}(R, B; \mathfrak{a}_{m!/(1-t),0}) \leq \frac{1-t}{m!} \operatorname{fpt}(R, B; \mathfrak{a}_{\bullet,0}) = \frac{\alpha}{m!} < \frac{x}{m!}.$$

Thus $(R, B, \mathfrak{a}_{m!/(1-t),0}^{x/m!})$ is not strongly F-regular by Definition 2.4, so $\tau_b(R, B, \mathfrak{a}_{m!/(1-t),0}^{x/m!}) \subset \mathfrak{m}_R$ by Definition-Theorem 2.3(1), and hence $\tau_b(R, B, \mathfrak{c}_0^{u_0} \cdots \mathfrak{c}_m^{u_m}) \subset \mathfrak{m}_R$.

Suppose that $y > \beta$. For any $0 \leq i \leq m-1$, Definition 3.1(3) implies

$$\mathfrak{c}_i^{m!/(m-i)} \subset \mathfrak{a}_{(m-i)/t,1}^{m!/(m-i)} \subset \mathfrak{a}_{m!/t,1},$$

and $\sum_{i=0}^{m-1}(m-i)u_i/m! = y/m!$. Since $\mathfrak{c}_m^{\lceil u_m q \rceil} \subset R$ for any $q = p^e$, Definition-Theorem 2.3(2) with c and $h = 1$ on both sides, and Lemma 3.7(2) applied to the ideals $\mathfrak{c}_0, \ldots, \mathfrak{c}_{m-1}$, the ideal $\mathfrak{a}_{m!/t,1}$, and the integers $m!/m, \ldots, m!/1$, imply

$$\tau_b(R, B, \mathfrak{c}_0^{u_0} \cdots \mathfrak{c}_m^{u_m}) \subset \tau_b(R, B, \mathfrak{c}_0^{u_0} \cdots \mathfrak{c}_{m-1}^{u_{m-1}}) \subset \tau_b(R, B, \mathfrak{a}_{m!/t,1}^{y/m!}).$$

By the supremum formula in Definition-Proposition 3.4 with $\lambda = m!/t$,

$$\operatorname{fpt}(R, B; \mathfrak{a}_{m!/t,1}) \leq \frac{t}{m!} \operatorname{fpt}(R, B; \mathfrak{a}_{\bullet,1}) = \frac{\beta}{m!} < \frac{y}{m!}.$$

Thus $(R, B, \mathfrak{a}_{m!/t,1}^{y/m!})$ is not strongly F-regular by Definition 2.4, so $\tau_b(R, B, \mathfrak{a}_{m!/t,1}^{y/m!}) \subset \mathfrak{m}_R$ by Definition-Theorem 2.3(1), and hence $\tau_b(R, B, \mathfrak{c}_0^{u_0} \cdots \mathfrak{c}_m^{u_m}) \subset \mathfrak{m}_R$.

Therefore any summand in the expansion of $\tau_b(R, B, \mathfrak{d}_m^u)$ above is contained in $\mathfrak{m}_R$, and so $\tau_b(R, B, \mathfrak{d}_m^u) \subset \mathfrak{m}_R$. By Definition-Theorem 2.3(1), $(R, B, \mathfrak{d}_m^u)$ is not strongly F-regular. Since $u > (\alpha+\beta)/m$ is arbitrary, Definition 2.4 implies $m \operatorname{fpt}(R, B; \mathfrak{d}_m) \leq \alpha + \beta$.

Step 3. In this step, we finish the proof.

By Lemma 3.2, $\mathfrak{a}_{\bullet,t}$ is a linearly bounded $\mathfrak{m}_R$-filtration. By **Steps 1 and 2** and Lemma 2.6(3), for any integer $m \geq 1$,

$$(m+1)\operatorname{fpt}(R, B; \mathfrak{a}_{m+1,t}) \leq (m+1)\operatorname{fpt}(R, B; \mathfrak{d}_m) \leq \frac{m+1}{m}(\alpha+\beta).$$

Thus by Definition-Proposition 3.4,

$$\begin{aligned}\operatorname{fpt}(R, B; \mathfrak{a}_{\bullet,t}) &= \lim_{m \to +\infty} (m+1)\operatorname{fpt}(R, B; \mathfrak{a}_{m+1,t}) \leq \alpha + \beta \\ &= (1-t)\operatorname{fpt}(R, B; \mathfrak{a}_{\bullet,0}) + t \operatorname{fpt}(R, B; \mathfrak{a}_{\bullet,1}).\end{aligned}$$

□

4. Intersections of filtrations and invariant ideals

In this section, we prove Theorem 4.1, that is, the F-volume can be computed by the ideals invariant under a finite group of automorphisms preserving the boundary. For a Galois cover with Galois group G, Proposition 5.5 only descends G-invariant ideals, so Proposition 5.5 and Theorem 4.1 together imply Theorem 5.2.

Theorem 4.1. *Notation and conditions as in Set-up 3.3. Let G be a finite group of ring automorphisms of R such that $g(B) = B$ for any $g \in G$. Then the following hold.*

(1) We have

$$\operatorname{Fvol}(R, B) = \inf_{\mathfrak{b}} e_R(\mathfrak{b}) \operatorname{fpt}(R, B; \mathfrak{b})^d,$$

where $\mathfrak{b}$ runs over the $\mathfrak{m}_R$-primary ideals of R such that $g(\mathfrak{b}) = \mathfrak{b}$ for any $g \in G$.

(2) Let $\mathfrak{a}$ be an $\mathfrak{m}_R$-primary ideal, $s := \operatorname{fpt}(R, B; \mathfrak{a})$, $u := e_R(\mathfrak{a})s^d$, and $\mathfrak{b}_m := \bigcap_{g \in G} g(\mathfrak{a}^{\lceil sm \rceil})$ for any positive integer m. Then $\mathfrak{b}_m$ is an $\mathfrak{m}_R$-primary ideal such that $g(\mathfrak{b}_m) = \mathfrak{b}_m$ for any $g \in G$, and for any real number $\epsilon > 0$, there exists a positive integer m_0 such that

$$e_R(\mathfrak{b}_m) \operatorname{fpt}(R, B; \mathfrak{b}_m)^d \leq u + (|G|-1)\sqrt{u^2 - \operatorname{Fvol}(R, B)^2} + \epsilon$$

for any integer $m \geq m_0$.

4.1. **Multiplicities of intersections.** In this subsection, we bound the multiplicities of the intersections of linearly bounded $\mathfrak{m}_R$-filtrations.

For linearly bounded $\mathfrak{m}_R$-filtrations $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$, the multiplicity of $\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}$ is at most the sum of the multiplicities of $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$, and Proposition 4.2 improves the bound by means of the multiplicity of $\mathfrak{a}_{\bullet,1/2}$. Proposition 4.2 and Corollary 4.3 assume that R is a Noetherian local domain containing a field such that $\widehat{R}$ is a domain, since the multiplicities of $\mathfrak{a}_{\bullet,0}$, $\mathfrak{a}_{\bullet,1/2}$, $\mathfrak{a}_{\bullet,1}$, and $\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}$ are the masses of different regions with respect to one measure [BLQ24, Proposition 4.7], which is constructed in [BLQ24, Section 4] for such R. For the proof of Theorem 4.1, a lower bound for the multiplicity of $\mathfrak{a}_{\bullet,1/2}$ follows from Theorem 3.6 and Proposition 3.5(1).

Proposition 4.2. *Let $(R, \mathfrak{m}_R)$ be a Noetherian local domain containing a field with $\dim R = d \geq 1$ such that $\widehat{R}$ is a domain. Let $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$ be linearly bounded $\mathfrak{m}_R$-filtrations, and let $u := e(\mathfrak{a}_{\bullet,0})$, $v := e(\mathfrak{a}_{\bullet,1})$, and $w := e(\mathfrak{a}_{\bullet,1/2})$. Then $0 < w \leq \sqrt{uv}$ and*

$$\max\{u, v\} \leq e(\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}) \leq \frac{u + v + \sqrt{(u+v)^2 - 4w^2}}{2}.$$

In particular, if $u = v$, then

$$0 \leq e(\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}) - u \leq \sqrt{u^2 - w^2}.$$

Proof. By Lemma 3.2, $\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}$ and $\mathfrak{a}_{\bullet,1/2}$ are linearly bounded $\mathfrak{m}_R$-filtrations. Since $\widehat{R}$ is a domain, hence reduced, the multiplicity of any $\mathfrak{m}_R$-filtration exists [Cut15, Theorem 6.5]. Let $\mathfrak{a}_\bullet$ be a linearly bounded $\mathfrak{m}_R$-filtration, and let $b > 0$ be a real number such that $\mathfrak{a}_\lambda \subset \mathfrak{m}_R^{\lceil b\lambda \rceil}$ for any $\lambda > 0$. Since $\mathfrak{a}_1^m \subset \mathfrak{a}_m \subset \mathfrak{m}_R^{\lceil bm \rceil}$ for any positive integer m, and since $\ell_R(R/\mathfrak{m}_R^n)$ agrees with a polynomial in n of degree d for any sufficiently large integer n [HS06, Theorem 11.1.3], we have

$$0 < b^d e_R(\mathfrak{m}_R) \leq e(\mathfrak{a}_\bullet) \leq e_R(\mathfrak{a}_1) < +\infty.$$

In particular, u, v, w, and $e(\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1})$ are positive real numbers.

Step 1. In this step, we express u, v, and w as integrals with respect to a finite measure ρ on $[0, 1]$.

Since $\widehat{R}$ is a domain, R contains a field, and u and v are positive real numbers, let μ be the measure on $\mathbb{R}^2$ of [BLQ24, Proposition 4.6] associated with $\mathfrak{a}_{\bullet,0}$ and $\mathfrak{a}_{\bullet,1}$. By [BLQ24, Proposition 4.9], μ is supported in $\mathbb{R}^2_{\geq 0}$ and homogeneous of degree d, that is, $\mu(sU) = s^d \mu(U)$ for any Borel set $U \subset \mathbb{R}^2_{\geq 0}$ and any real number $s > 0$. We write subsets of $\mathbb{R}^2_{\geq 0}$ in the coordinates (x, y), for example, $\{x < 1\} := \{(x, y) \in \mathbb{R}^2_{\geq 0} \mid x < 1\}$. By [BLQ24, Proposition 4.7],

$$\mu(\{(1-t)x + ty < 1\}) = e(\mathfrak{a}_{\bullet,t}) \quad \text{for any } t \in [0, 1], \tag{4.1}$$

$$\mu(\{x < 1 \text{ or } y < 1\}) = e(\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}). \tag{4.2}$$

For any real number $s > 0$, homogeneity and (4.1) with $t = 0$ and $t = 1$ imply

$$\mu(\{x = 0\}) \leq \mu(\{x < s\}) = s^d u \quad \text{and} \quad \mu(\{y = 0\}) \leq \mu(\{y < s\}) = s^d v.$$

Letting $s \to 0$, we obtain $\mu(\{x = 0\}) = \mu(\{y = 0\}) = 0$. In particular, $\mu(\{(0, 0)\}) = 0$.

Let ρ be the measure on $[0, 1]$ of [BLQ24, Section 4.4], that is,

$$\rho(U) := \mu(\{0 < x + y < 1 \text{ and } x/(x+y) \in U\})$$

for any Borel set $U \subset [0, 1]$. Then $\rho([0, 1]) \leq \mu(\{x < 1\}) = u < +\infty$. Moreover, we have $\rho(\{0\}) \leq \mu(\{x = 0\}) = 0$ and $\rho(\{1\}) \leq \mu(\{y = 0\}) = 0$. Let $\varphi_0(z) := z^{-d}$ and $\varphi_1(z) := (1-z)^{-d}$ for $z \in (0, 1)$. In the rest of the proof, integrals with respect to ρ are over $(0, 1)$. By [BLQ24, Lemma 4.11] with $t = 0$, $1/2$, and 1, we have

$$u = \int \varphi_0 d\rho, \quad w = 2^d \rho((0, 1)), \quad \text{and} \quad v = \int \varphi_1 d\rho. \tag{4.3}$$

Step 2. In this step, we prove that

$$e(\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}) = \int \max\{\varphi_0, \varphi_1\} d\rho. \tag{4.4}$$

Let $U \subset [0,1]$ be a Borel set and $s > 0$ a real number. Since $\{0 < x+y < s \text{ and } x/(x+y) \in U\}$ is the image of $\{0 < x+y < 1 \text{ and } x/(x+y) \in U\}$ under multiplication by s, homogeneity implies

$$\mu(\{0 < x+y < s \text{ and } x/(x+y) \in U\}) = s^d \rho(U). \tag{4.5}$$

Let η be the Lebesgue–Stieltjes measure on $(0, +\infty)$ of the function s^d, that is, $\eta((0,s)) = s^d$ for any real number $s > 0$. Let

$$\Psi : \mathbb{R}^2_{\geq 0} \backslash \{(0,0)\} \to (0, +\infty) \times [0,1]$$

be the homeomorphism sending (x,y) to $(x+y, x/(x+y))$, and let $\Psi_* \mu$ be the push-forward of the restriction of μ to $\mathbb{R}^2_{\geq 0} \backslash \{(0,0)\}$. By (4.5), $\Psi_* \mu$ and the product measure $\eta \otimes \rho$ satisfy

$$\Psi_* \mu((0,s) \times U) = s^d \rho(U) = (\eta \otimes \rho)((0,s) \times U)$$

for any real number $s > 0$ and any Borel set $U \subset [0,1]$. The sets $(0,s) \times U$ for real numbers $s > 0$ and Borel sets $U \subset [0,1]$ form a π-system which generates the Borel σ-algebra of $(0, +\infty) \times [0,1]$, and $\Psi_* \mu((0,n) \times [0,1]) = n^d \rho([0,1]) < +\infty$ for any positive integer n. Thus $\Psi_* \mu = \eta \otimes \rho$ by Dynkin's π-λ theorem.

Let $\kappa(z) := \max\{1/z, 1/(1-z)\} \in (0, +\infty]$ for $z \in [0,1]$, where $1/0 := +\infty$. For $z := x/(x+y)$ with $(x,y) \in \mathbb{R}^2_{\geq 0} \backslash \{(0,0)\}$, we have $x = (x+y)z$ and $y = (x+y)(1-z)$. Thus $x < 1$ or $y < 1$ if and only if $x + y < \kappa(z)$. Since $\Psi_* \mu = \eta \otimes \rho$, the Tonelli theorem implies

$$\begin{aligned} \mu(\{x < 1 \text{ or } y < 1\} \backslash \{(0,0)\}) &= (\eta \otimes \rho)(\{(s,z) \mid s < \kappa(z)\}) \\ &= \int_{[0,1]} \eta((0, \kappa(z))) d\rho(z) = \int_{[0,1]} \kappa^d d\rho. \end{aligned}$$

Since $\mu(\{(0,0)\}) = 0$, $\rho(\{0\}) = \rho(\{1\}) = 0$, and $\kappa^d = \max\{\varphi_0, \varphi_1\}$ on $(0,1)$, (4.2) implies (4.4).

Step 3. In this step, we deduce the inequalities from (4.3) and (4.4) by the Cauchy–Schwarz inequality. Let

$$Q := \int \sqrt{\varphi_0 \varphi_1} d\rho \quad \text{and} \quad T := \int |\varphi_0 - \varphi_1| d\rho.$$

Since $z(1-z) \leq 1/4$ for any $z \in (0,1)$, we have $\sqrt{\varphi_0(z)\varphi_1(z)} = (z(1-z))^{-d/2} \geq 2^d$. Thus $Q \geq 2^d \rho((0,1)) = w$ by (4.3). By the Cauchy–Schwarz inequality and (4.3), $Q \leq \sqrt{uv}$. Thus $0 < w \leq Q \leq \sqrt{uv} < +\infty$, and

$$\int (\sqrt{\varphi_0} \pm \sqrt{\varphi_1})^2 d\rho = u + v \pm 2Q.$$

By the Cauchy–Schwarz inequality again,

$$\begin{aligned} T^2 &= \left(\int |\sqrt{\varphi_0} - \sqrt{\varphi_1}| (\sqrt{\varphi_0} + \sqrt{\varphi_1}) d\rho \right)^2 \\ &\leq \int (\sqrt{\varphi_0} - \sqrt{\varphi_1})^2 d\rho \cdot \int (\sqrt{\varphi_0} + \sqrt{\varphi_1})^2 d\rho \\ &= (u + v - 2Q)(u + v + 2Q) \\ &= (u+v)^2 - 4Q^2 \\ &\leq (u+v)^2 - 4w^2, \end{aligned}$$

where the last inequality follows from $Q \geq w > 0$. Therefore $T \leq \sqrt{(u+v)^2 - 4w^2}$. Since $\max\{\varphi_0, \varphi_1\} = (\varphi_0 + \varphi_1 + |\varphi_0 - \varphi_1|)/2$, (4.3) and (4.4) imply

$$e(\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}) = \frac{u + v + T}{2} \leq \frac{u + v + \sqrt{(u+v)^2 - 4w^2}}{2}.$$

Since $\max\{\varphi_0,\varphi_1\} \geq \varphi_i$ for $i \in \{0,1\}$, (4.3) and (4.4) also imply $e(\mathfrak{a}_{\bullet,0} \cap \mathfrak{a}_{\bullet,1}) \geq \max\{u,v\}$. Finally, if $u = v$, then the upper bound is $u + \sqrt{u^2 - w^2}$ and the lower bound is u. □

Corollary 4.3 bounds the multiplicity of the intersection of the translates of a linearly bounded $\mathfrak{m}_R$-filtration under a finite group.

Corollary 4.3. *Let $(R,\mathfrak{m}_R)$ be a Noetherian local domain containing a field with $\dim R = d \geq 1$ such that $\widehat{R}$ is a domain. Let G be a finite group of ring automorphisms of R, and $\mathfrak{a}_\bullet$ a linearly bounded $\mathfrak{m}_R$-filtration with $u := e(\mathfrak{a}_\bullet)$. For $g \in G$, let $g\mathfrak{a}_\bullet := (g(\mathfrak{a}_\lambda))_{\lambda>0}$, let $(\mathfrak{a}^g_{\bullet,t})_{t\in[0,1]}$ be the geodesic between $\mathfrak{a}_\bullet$ and $g\mathfrak{a}_\bullet$, and let $w_g := e(\mathfrak{a}^g_{\bullet,1/2})$. Let $\mathfrak{b}_\lambda := \bigcap_{g\in G} g(\mathfrak{a}_\lambda)$ for $\lambda > 0$. Then $\mathfrak{b}_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration such that $g(\mathfrak{b}_\lambda) = \mathfrak{b}_\lambda$ for any $g \in G$ and any $\lambda > 0$, and*

$$u \leq e(\mathfrak{b}_\bullet) \leq u + \sum_{g\in G\backslash\{1\}} \sqrt{u^2 - w_g^2}.$$

In particular, if v is a real number such that $0 \leq v \leq u$ and $w_g \geq v$ for any $g \in G\backslash\{1\}$, then $e(\mathfrak{b}_\bullet) \leq u + (|G|-1)\sqrt{u^2 - v^2}$.

Proof. Let $g \in G$. Since g is a ring automorphism of R, $g\mathfrak{a}_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration with $e(g\mathfrak{a}_\bullet) = e(\mathfrak{a}_\bullet) = u$.

By Lemma 3.2, $\mathfrak{a}_\bullet \cap g\mathfrak{a}_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration for any $g \in G$. By Lemma 3.2 applied $|G|-1$ times, $\mathfrak{b}_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration. Since $\widehat{R}$ is a domain, hence reduced, $e(\mathfrak{a}_\bullet \cap g\mathfrak{a}_\bullet)$ and $e(\mathfrak{b}_\bullet)$ exist [Cut15, Theorem 6.5]. For $h \in G$ and $\lambda > 0$, we have

$$h(\mathfrak{b}_\lambda) = \bigcap_{g\in G} (hg)(\mathfrak{a}_\lambda) = \mathfrak{b}_\lambda,$$

since left multiplication by h is a bijection of G.

Let $\lambda > 0$. Since $1 \in G$, we have $\mathfrak{b}_\lambda \subset \mathfrak{a}_\lambda \cap g(\mathfrak{a}_\lambda) \subset \mathfrak{a}_\lambda$ for any $g \in G$. The homomorphism of R-modules of finite length

$$\mathfrak{a}_\lambda/\mathfrak{b}_\lambda \to \bigoplus_{g\in G\backslash\{1\}} \mathfrak{a}_\lambda/(\mathfrak{a}_\lambda \cap g(\mathfrak{a}_\lambda))$$

sending the class of $y \in \mathfrak{a}_\lambda$ to the class of y in each summand is injective, since an element of $\mathfrak{a}_\lambda$ contained in $g(\mathfrak{a}_\lambda)$ for any $g \in G\backslash\{1\}$ is contained in $\mathfrak{b}_\lambda$. Taking lengths, we obtain

$$\ell_R(R/\mathfrak{b}_\lambda) - \ell_R(R/\mathfrak{a}_\lambda) \leq \sum_{g\in G\backslash\{1\}} \big(\ell_R(R/(\mathfrak{a}_\lambda \cap g(\mathfrak{a}_\lambda))) - \ell_R(R/\mathfrak{a}_\lambda)\big). \tag{4.6}$$

Applying (4.6) with $\lambda = m$ for any positive integer m, multiplying by $d!/m^d$, and letting $m \to +\infty$, we obtain

$$e(\mathfrak{b}_\bullet) - u \leq \sum_{g\in G\backslash\{1\}} \big(e(\mathfrak{a}_\bullet \cap g\mathfrak{a}_\bullet) - u\big).$$

For any $g \in G\backslash\{1\}$, by Proposition 4.2 applied to $\mathfrak{a}_\bullet$ and $g\mathfrak{a}_\bullet$, both of which have multiplicity u, we have $0 \leq e(\mathfrak{a}_\bullet \cap g\mathfrak{a}_\bullet) - u \leq \sqrt{u^2 - w_g^2}$. Moreover, $e(\mathfrak{b}_\bullet) \geq u$ since $\mathfrak{b}_\lambda \subset \mathfrak{a}_\lambda$ for any $\lambda > 0$. Thus

$$u \leq e(\mathfrak{b}_\bullet) \leq u + \sum_{g\in G\backslash\{1\}} \sqrt{u^2 - w_g^2}.$$

Finally, assume that $0 \leq v \leq u$ and $w_g \geq v$ for any $g \in G\backslash\{1\}$. Then $\sqrt{u^2 - w_g^2} \leq \sqrt{u^2 - v^2}$ for any $g \in G\backslash\{1\}$. Since $G\backslash\{1\}$ has $|G|-1$ elements, we obtain $e(\mathfrak{b}_\bullet) \leq u + (|G|-1)\sqrt{u^2 - v^2}$. □

4.2. **Invariant ideals.** In this subsection, we prove Theorem 4.1.

Proof of Theorem 4.1. We first prove (2), and then deduce (1) from (2). Recall that in (2), $\mathfrak{a}$ is an $\mathfrak{m}_R$-primary ideal, $s = \operatorname{fpt}(R, B; \mathfrak{a})$, $u = e_R(\mathfrak{a})s^d$, and $\mathfrak{b}_m = \bigcap_{g\in G} g(\mathfrak{a}^{\lceil sm\rceil})$ for any positive integer m.

Let $\mathfrak{a}'_\lambda := \mathfrak{a}^{\lceil s\lambda\rceil}$ for $\lambda > 0$. By Proposition 3.5(2), $\mathfrak{a}'_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration with $\operatorname{fpt}(R, B; \mathfrak{a}'_\bullet) = 1$ and $e(\mathfrak{a}'_\bullet) = u$. For $g \in G$, let $g\mathfrak{a}'_\bullet := (g(\mathfrak{a}'_\lambda))_{\lambda>0}$. Since $g(\mathfrak{m}_R) = \mathfrak{m}_R$, $g\mathfrak{a}'_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration.

We show that $\operatorname{fpt}(R, B; g\mathfrak{a}'_\bullet) = 1$ for any $g \in G$. Let $g \in G$. Since g commutes with the Frobenius morphism and $g(B) = B$, conjugating by g the maps in Definition 2.2 shows that if $(R, B, \mathfrak{c}^\gamma)$ is strongly F-regular for a non-zero ideal $\mathfrak{c} \subset R$ and a real number $\gamma \geq 0$, then $(R, B, g(\mathfrak{c})^\gamma)$ is strongly F-regular. Replacing g by $g^{-1} \in G$ and $\mathfrak{c}$ by $g(\mathfrak{c})$ in the last implication, we obtain the converse, hence

$$\operatorname{fpt}(R, B; g(\mathfrak{c})) = \operatorname{fpt}(R, B; \mathfrak{c}) \quad \text{for any } g \in G \text{ and any non-zero ideal } \mathfrak{c} \subset R. \tag{4.7}$$

Taking $\mathfrak{c} = \mathfrak{a}'_\lambda$ for $\lambda > 0$ in (4.7), Definition-Proposition 3.4 implies

$$\operatorname{fpt}(R, B; g\mathfrak{a}'_\bullet) = \lim_{m\to+\infty} m \operatorname{fpt}(R, B; g(\mathfrak{a}'_m)) = \lim_{m\to+\infty} m \operatorname{fpt}(R, B; \mathfrak{a}'_m) = \operatorname{fpt}(R, B; \mathfrak{a}'_\bullet) = 1.$$

For $g \in G$, let $(\mathfrak{a}^g_{\bullet,t})_{t\in[0,1]}$ be the geodesic between $\mathfrak{a}'_\bullet$ and $g\mathfrak{a}'_\bullet$, and let $w_g := e(\mathfrak{a}^g_{\bullet,1/2})$. By Lemma 3.2, $\mathfrak{a}^g_{\bullet,1/2}$ is a linearly bounded $\mathfrak{m}_R$-filtration. By Theorem 3.6,

$$\operatorname{fpt}(R, B; \mathfrak{a}^g_{\bullet,1/2}) \leq \frac{1}{2}\operatorname{fpt}(R, B; \mathfrak{a}'_\bullet) + \frac{1}{2}\operatorname{fpt}(R, B; g\mathfrak{a}'_\bullet) = 1.$$

Thus Proposition 3.5(1) implies

$$\operatorname{Fvol}(R, B) \leq w_g \operatorname{fpt}(R, B; \mathfrak{a}^g_{\bullet,1/2})^d \leq w_g.$$

Moreover, $0 \leq \operatorname{Fvol}(R, B) \leq e_R(\mathfrak{a})s^d = u$ by Definition 2.4.

Let $\mathfrak{b}_\lambda := \bigcap_{g\in G} g(\mathfrak{a}'_\lambda)$ for $\lambda > 0$. For any positive integer m, $\mathfrak{b}_m = \bigcap_{g\in G} g(\mathfrak{a}^{\lceil sm\rceil})$ coincides with the ideal $\mathfrak{b}_m$ in (2). Since R is excellent and normal, $\widehat{R}$ is normal [Gro65, (7.8.3)(v)], hence $\widehat{R}$ is a domain. Moreover, R contains $\mathbb{F}_p$. Thus Corollary 4.3 with $v = \operatorname{Fvol}(R, B)$ implies that $\mathfrak{b}_\bullet$ is a linearly bounded $\mathfrak{m}_R$-filtration such that $g(\mathfrak{b}_\lambda) = \mathfrak{b}_\lambda$ for any $g \in G$ and any $\lambda > 0$, and

$$e(\mathfrak{b}_\bullet) \leq u + (|G| - 1)\sqrt{u^2 - \operatorname{Fvol}(R, B)^2}.$$

In particular, $\mathfrak{b}_m$ is an $\mathfrak{m}_R$-primary ideal such that $g(\mathfrak{b}_m) = \mathfrak{b}_m$ for any $g \in G$.

Since $\mathfrak{b}_\lambda \subset \mathfrak{a}'_\lambda$ for any $\lambda > 0$, Definition-Proposition 3.4(2) and the supremum formula in Definition-Proposition 3.4 imply

$$m \operatorname{fpt}(R, B; \mathfrak{b}_m) \leq \sup_{m'\geq 1} m' \operatorname{fpt}(R, B; \mathfrak{b}_{m'}) = \operatorname{fpt}(R, B; \mathfrak{b}_\bullet) \leq \operatorname{fpt}(R, B; \mathfrak{a}'_\bullet) = 1$$

for any positive integer m. Thus

$$e_R(\mathfrak{b}_m) \operatorname{fpt}(R, B; \mathfrak{b}_m)^d \leq \frac{e_R(\mathfrak{b}_m)}{m^d}$$

for any positive integer m. By Definition-Proposition 3.4(1), $e_R(\mathfrak{b}_m)/m^d$ converges to $e(\mathfrak{b}_\bullet)$ as $m \to +\infty$. Thus for any real number $\epsilon > 0$, there exists a positive integer m_0 such that

$$e_R(\mathfrak{b}_m) \operatorname{fpt}(R, B; \mathfrak{b}_m)^d \leq e(\mathfrak{b}_\bullet) + \epsilon \leq u + (|G| - 1)\sqrt{u^2 - \operatorname{Fvol}(R, B)^2} + \epsilon$$

for any integer $m \geq m_0$. This proves (2).

We prove (1). Since the infimum in (1) runs over a subset of the $\mathfrak{m}_R$-primary ideals of R, it is at least $\operatorname{Fvol}(R, B)$. Let $\epsilon > 0$ be a real number. By Definition 2.4, there exists an $\mathfrak{m}_R$-primary ideal $\mathfrak{a}$ such that $u = e_R(\mathfrak{a}) \operatorname{fpt}(R, B; \mathfrak{a})^d \leq \operatorname{Fvol}(R, B) + \epsilon$. Then

$$u^2 - \operatorname{Fvol}(R, B)^2 = (u - \operatorname{Fvol}(R, B))(u + \operatorname{Fvol}(R, B)) \leq \epsilon(2 \operatorname{Fvol}(R, B) + \epsilon).$$

By (2), there exists a positive integer m such that $\mathfrak{b}_m$ is an $\mathfrak{m}_R$-primary ideal with $g(\mathfrak{b}_m) = \mathfrak{b}_m$ for any $g \in G$ and

$$\begin{aligned} e_R(\mathfrak{b}_m)\operatorname{fpt}(R, B; \mathfrak{b}_m)^d &\leq u + (|G|-1)\sqrt{u^2 - \operatorname{Fvol}(R,B)^2} + \epsilon \\ &\leq \operatorname{Fvol}(R,B) + 2\epsilon + (|G|-1)\sqrt{\epsilon(2\operatorname{Fvol}(R,B)+\epsilon)}. \end{aligned}$$

Letting $\epsilon \to 0$, we see that the infimum in (1) is at most $\operatorname{Fvol}(R, B)$. This proves (1). □

5. Galois covers of complete local domains

The goal of this section is to prove Theorem 5.2. By Theorem 4.1, the F-volume of the larger ring can be computed using invariant ideals only, and we descend such ideals to the smaller ring after raising them to a power that depends only on the order of the Galois group.

Set-up 5.1. Let R, S, d, G, B, and B_S be as follows.

(1) $(R, \mathfrak{m}_R) \subset (S, \mathfrak{m}_S)$ is a module-finite inclusion of complete Noetherian normal local domains containing k, with $R/\mathfrak{m}_R = S/\mathfrak{m}_S = k$ and $\dim R = \dim S = d \geq 1$.
(2) $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale in codimension one.
(3) $\operatorname{Frac} S / \operatorname{Frac} R$ is a Galois extension with group G.
(4) $B \geq 0$ is a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$ such that (R, B) is strongly F-regular.
(5) B_S is the divisorial pullback of B to $\operatorname{Spec} S$.

Since R and S are complete local domains containing the perfect field k with residue field k, they are F-finite by the Cohen structure theorem [HS06, Theorem 4.3.3]. Thus condition (4) of Set-up 5.1 and the strong F-regularity of pairs on $\operatorname{Spec} S$ in this section are within the scope of Definition 2.2. By Proposition 5.5(1) below, the pair (S, B_S) is strongly F-regular, so $\operatorname{Fvol}(S, B_S)$ is defined.

Theorem 5.2. *Notation and conditions as in Set-up 5.1. Then*

$$\operatorname{Fvol}(S, B_S) = |G| \cdot \operatorname{Fvol}(R, B).$$

In general, G-invariant ideals of S are not extended from ideals of R (Example 5.4). In the following lemma, we compare the powers of a G-invariant ideal with the powers of an ideal extended from R. The inclusions in the lemma suffice for the computation of multiplicities and F-pure thresholds in Proposition 5.5.

Lemma 5.3. *Let S be a Noetherian local domain and G a finite group of ring automorphisms of S such that $R := S^G$ is Noetherian. Let $\mathfrak{b} \subset S$ be an ideal such that $g(\mathfrak{b}) = \mathfrak{b}$ for any $g \in G$, let $N := \operatorname{lcm}(1, 2, \ldots, |G|)$, and let $\mathfrak{a} := \mathfrak{b}^N \cap R$. Then any element of $\mathfrak{b}^N$ is integral over the ideal $\mathfrak{a}S$, and there exists a positive integer h such that*

$$\mathfrak{b}^{N(m+h)} \subset (\mathfrak{a}S)^m \subset \mathfrak{b}^{Nm}$$

for any positive integer m.

Proof. We refer to [HS06, Definition 1.1.1] for elements integral over an ideal. Let w be an indeterminate over S, and let

$$S[\mathfrak{b}w] := \bigoplus_{n \geq 0} \mathfrak{b}^n w^n \subset S[w]$$

be the Rees algebra of $\mathfrak{b}$. Let

$$S[\mathfrak{a}w^N] := \bigoplus_{n \geq 0} (\mathfrak{a}S)^n w^{Nn}.$$

Since $\mathfrak{a}S \subset \mathfrak{b}^N$, $S[\mathfrak{a}w^N]$ is a graded subring of $S[\mathfrak{b}w]$.

Let $f \in \mathfrak{b}$, and write

$$\prod_{g \in G} (T - g(f)) = T^{|G|} + c_1 T^{|G|-1} + \cdots + c_{|G|}$$

in the polynomial ring $S[T]$. For any $g' \in G$, we have $\prod_{g\in G}(T - g'(g(f))) = \prod_{g\in G}(T - g(f))$, so $g'(c_i) = c_i$ for any i. Thus $c_i \in S^G = R$. Since c_i is, up to sign, the i-th elementary symmetric polynomial in the elements $g(f) \in \mathfrak{b}$ with $g \in G$, we have $c_i \in \mathfrak{b}^i$. Since N/i is a positive integer for $1 \le i \le |G|$, we obtain

$$c_i^{N/i} \in \mathfrak{b}^N \cap R = \mathfrak{a}.$$

Thus $c_i w^i$ is a root of the monic polynomial $U^{N/i} - c_i^{N/i} w^N \in S[\mathfrak{a}w^N][U]$, so $c_i w^i$ is integral over $S[\mathfrak{a}w^N]$. Since the factor $T - f$ vanishes at $T = f$, substituting $T = f$ and multiplying by $w^{|G|}$, we obtain

$$(fw)^{|G|} + (c_1 w)(fw)^{|G|-1} + \cdots + c_{|G|} w^{|G|} = 0.$$

Thus fw is integral over the ring

$$S[\mathfrak{a}w^N][c_1 w, c_2 w^2, \ldots, c_{|G|} w^{|G|}],$$

which is generated over $S[\mathfrak{a}w^N]$ by finitely many elements integral over $S[\mathfrak{a}w^N]$. Let $f_1, \ldots, f_s$ be generators of $\mathfrak{b}$. Since $S[\mathfrak{b}w] = S[\mathfrak{a}w^N][f_1 w, \ldots, f_s w]$, the ring $S[\mathfrak{b}w]$ is a finite $S[\mathfrak{a}w^N]$-module by [HS06, Lemma 2.1.9 and Corollary 2.1.12].

Let $z \in \mathfrak{b}^N$. Since zw^N belongs to the finite $S[\mathfrak{a}w^N]$-module $S[\mathfrak{b}w]$, the element zw^N is integral over $S[\mathfrak{a}w^N]$. Moreover, $S[\mathfrak{a}w^N]$ is the Rees algebra of $\mathfrak{a}S$ in the variable w^N over S, and $zw^N \in S[w^N]$. Thus z is integral over $\mathfrak{a}S$ by [HS06, Proposition 5.2.1].

Since $\mathfrak{a}S \subset \mathfrak{b}^N$, the ideal $\mathfrak{b}^N$ is finitely generated, and any element of $\mathfrak{b}^N$ is integral over $\mathfrak{a}S$, there exists a positive integer h such that $\mathfrak{b}^{N(m+h)} \subset (\mathfrak{a}S)^m$ for any positive integer m [HS06, Corollary 1.2.5 and Remark 1.2.3]. Moreover, $(\mathfrak{a}S)^m \subset \mathfrak{b}^{Nm}$ since $\mathfrak{a}S \subset \mathfrak{b}^N$. □

Example 5.4. Assume that $p \neq 2$. Let $S := k[[u, v]]$, let σ be the k-algebra automorphism of S such that $\sigma(u) = -u$ and $\sigma(v) = -v$, let $G := \{1, \sigma\}$, and let $R := S^G$. Then R consists of the power series in S whose monomials have even degree, so $R = k[[u^2, uv, v^2]]$. Moreover, $N = 2$ in Lemma 5.3. An ideal $\mathfrak{c} \subset S$ is the extension of an ideal of R if and only if $\mathfrak{c} = (\mathfrak{c} \cap R)S$: if $\mathfrak{c} = \mathfrak{c}_0 S$ for an ideal $\mathfrak{c}_0 \subset R$, then $\mathfrak{c}_0 \subset \mathfrak{c} \cap R$, so $\mathfrak{c} = \mathfrak{c}_0 S \subset (\mathfrak{c} \cap R)S \subset \mathfrak{c}$.

(1) We have $\sigma(\mathfrak{m}_S) = \mathfrak{m}_S$ and $\mathfrak{m}_S \cap R = \mathfrak{m}_R = (u^2, uv, v^2)R$. Thus $(\mathfrak{m}_S \cap R)S = \mathfrak{m}_S^2 \neq \mathfrak{m}_S$, so $\mathfrak{m}_S$ is not the extension of an ideal of R.

(2) Let $\mathfrak{b} := (u, v^2)$. Then $\sigma(\mathfrak{b}) = \mathfrak{b}$ and $\mathfrak{b}^2 = (u^2, uv^2, v^4)$. Since $\mathfrak{b}^2$ is generated by monomials, a direct calculation shows that $(\mathfrak{b}^2 \cap R)S = (u^2, uv^3, v^4)$. Thus $uv^2 \in \mathfrak{b}^2$ does not belong to $(\mathfrak{b}^2 \cap R)S$, while $(uv^2)^2 = u^2 \cdot v^4 \in ((\mathfrak{b}^2 \cap R)S)^2$. Hence the integral dependence in Lemma 5.3 cannot be replaced by the equality $\mathfrak{b}^N = (\mathfrak{b}^N \cap R)S$.

Proposition 5.5. *Notation and conditions as in Set-up 5.1. Let $N := \operatorname{lcm}(1, 2, \ldots, |G|)$. Then the following hold.*

(1) The group G acts faithfully on S by ring automorphisms with $S^G = R$, $g(B_S) = B_S$ for any $g \in G$, the pair (S, B_S) is strongly F-regular, and $\operatorname{fpt}(R, B; \mathfrak{a}) = \operatorname{fpt}(S, B_S; \mathfrak{a}S)$ for any non-zero ideal $\mathfrak{a} \subset R$.

(2) Let $\mathfrak{b}$ be an $\mathfrak{m}_S$-primary ideal such that $g(\mathfrak{b}) = \mathfrak{b}$ for any $g \in G$, and let $\mathfrak{a} := \mathfrak{b}^N \cap R$. Then $\mathfrak{a}$ is $\mathfrak{m}_R$-primary,

$$e_R(\mathfrak{a}) = \frac{N^d}{|G|} e_S(\mathfrak{b}), \quad \operatorname{fpt}(R, B; \mathfrak{a}) = \frac{\operatorname{fpt}(S, B_S; \mathfrak{b})}{N},$$

and

$$|G| \cdot e_R(\mathfrak{a}) \operatorname{fpt}(R, B; \mathfrak{a})^d = e_S(\mathfrak{b}) \operatorname{fpt}(S, B_S; \mathfrak{b})^d.$$

(3) We have

$$\inf_{\mathfrak{b}} e_S(\mathfrak{b}) \operatorname{fpt}(S, B_S; \mathfrak{b})^d = |G| \cdot \operatorname{Fvol}(R, B),$$

where $\mathfrak{b}$ runs over the $\mathfrak{m}_S$-primary ideals of S such that $g(\mathfrak{b}) = \mathfrak{b}$ for any $g \in G$.

Proof. (1) Since S is normal and module-finite over R, S is the integral closure of R in $\operatorname{Frac} S$. Let $g \in G$. Since g is an automorphism of $\operatorname{Frac} S$ fixing $\operatorname{Frac} R$ pointwise, g maps elements of $\operatorname{Frac} S$ integral over R to elements integral over R. Thus $g(S) \subset S$. Since $g^{-1} \in G$, we also have $g^{-1}(S) \subset S$, so $g(S) = S$. Since any element of $\operatorname{Frac} S$ is a quotient of two elements of S, an element of G fixing S pointwise is the identity. Thus G acts faithfully on S. Since $(\operatorname{Frac} S)^G = \operatorname{Frac} R$, S is integral over R, and R is normal, we have

$$S^G = S \cap (\operatorname{Frac} S)^G = S \cap \operatorname{Frac} R = R.$$

Next, we show that $g(B_S) = B_S$. By Definition 2.1,

$$g(B_S) = \sum_Q b_{Q\cap R} \operatorname{ord}_Q(t_{Q\cap R}) g(Q),$$

where Q runs over the height-one primes of S. Let Q be a height-one prime of S. Then g induces an isomorphism of discrete valuation rings $S_Q \to S_{g(Q)}$, so $\operatorname{ord}_{g(Q)}(g(z)) = \operatorname{ord}_Q(z)$ for any non-zero $z \in \operatorname{Frac} S$. Since g fixes $\operatorname{Frac} R$ pointwise, we have $g(Q) \cap R = Q \cap R$ and $g(t_{Q\cap R}) = t_{Q\cap R}$. Thus the coefficient of B_S at $g(Q)$ is

$$b_{g(Q)\cap R} \operatorname{ord}_{g(Q)}(t_{g(Q)\cap R}) = b_{Q\cap R} \operatorname{ord}_{g(Q)}(g(t_{Q\cap R})) = b_{Q\cap R} \operatorname{ord}_Q(t_{Q\cap R}),$$

which is the coefficient of $g(B_S)$ at $g(Q)$. Since g permutes the height-one primes of S, we obtain $g(B_S) = B_S$.

Finally, R and S are F-finite by the Cohen structure theorem [HS06, Theorem 4.3.3], since k is perfect. Moreover, R is strongly F-regular by Lemma 2.6(1). Since R and S are complete and $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale in codimension one, Theorem 2.7(3) implies that (S, B_S) is strongly F-regular and $\operatorname{fpt}(R, B; \mathfrak{a}) = \operatorname{fpt}(S, B_S; \mathfrak{a}S)$ for any non-zero ideal $\mathfrak{a} \subset R$.

(2) By (1), G is a finite group of ring automorphisms of S with $S^G = R$. Since S is a Noetherian local domain and R is Noetherian, Lemma 5.3 implies that any element of $\mathfrak{b}^N$ is integral over $\mathfrak{a}S$, and that there exists a positive integer h such that

$$\mathfrak{b}^{N(m+h)} \subset (\mathfrak{a}S)^m \subset \mathfrak{b}^{Nm} \tag{5.1}$$

for any positive integer m.

Since radicals commute with contraction and $\mathfrak{b}$ is $\mathfrak{m}_S$-primary, we have $\sqrt{\mathfrak{a}} = \mathfrak{m}_S \cap R$. Since $\mathfrak{m}_S \cap R = \mathfrak{m}_R$ [HS06, Lemma 2.1.7], $\mathfrak{a}$ is $\mathfrak{m}_R$-primary, and $\mathfrak{a} \neq 0$ since $d \geq 1$. Moreover, $\mathfrak{a}S \subset \mathfrak{m}_S$, and $\mathfrak{a}S$ contains the $\mathfrak{m}_S$-primary ideal $\mathfrak{b}^{N(1+h)}$ by (5.1). Thus $\mathfrak{a}S$ is $\mathfrak{m}_S$-primary.

By (1), (S, B_S) is strongly F-regular. Thus, by Lemma 2.6(3) and (4) for (S, B_S), (5.1) implies that

$$\frac{\operatorname{fpt}(S, B_S; \mathfrak{b})}{N(m+h)} \leq \frac{\operatorname{fpt}(S, B_S; \mathfrak{a}S)}{m} \leq \frac{\operatorname{fpt}(S, B_S; \mathfrak{b})}{Nm}$$

for any positive integer m. Letting $m \to +\infty$, we obtain $\operatorname{fpt}(S, B_S; \mathfrak{a}S) = \operatorname{fpt}(S, B_S; \mathfrak{b})/N$. By (1), we have

$$\operatorname{fpt}(R, B; \mathfrak{a}) = \operatorname{fpt}(S, B_S; \mathfrak{a}S) = \frac{\operatorname{fpt}(S, B_S; \mathfrak{b})}{N}.$$

Since $\mathfrak{a}S \subset \mathfrak{b}^N$ and any element of $\mathfrak{b}^N$ is integral over $\mathfrak{a}S$, we have

$$e_S(\mathfrak{a}S) = e_S(\mathfrak{b}^N) = N^d e_S(\mathfrak{b})$$

by [HS06, Propositions 11.2.1 and 11.2.9(1)]. Since S is local, $R/\mathfrak{m}_R = S/\mathfrak{m}_S$, and the degree $[\operatorname{Frac} S : \operatorname{Frac} R]$ equals $|G|$, we have $e_S(\mathfrak{a}S) = |G| e_R(\mathfrak{a})$ by [HS06, Theorem 11.2.7]. Thus $e_R(\mathfrak{a}) = N^d e_S(\mathfrak{b})/|G|$, and

$$|G| \cdot e_R(\mathfrak{a}) \operatorname{fpt}(R, B; \mathfrak{a})^d = N^d e_S(\mathfrak{b}) \cdot \frac{\operatorname{fpt}(S, B_S; \mathfrak{b})^d}{N^d} = e_S(\mathfrak{b}) \operatorname{fpt}(S, B_S; \mathfrak{b})^d.$$

(3) Let $\mathfrak{b}$ be an $\mathfrak{m}_S$-primary ideal such that $g(\mathfrak{b}) = \mathfrak{b}$ for any $g \in G$. By (2), the ideal $\mathfrak{b}^N \cap R$ is $\mathfrak{m}_R$-primary and

$$e_S(\mathfrak{b}) \operatorname{fpt}(S, B_S; \mathfrak{b})^d = |G| \cdot e_R(\mathfrak{b}^N \cap R) \operatorname{fpt}(R, B; \mathfrak{b}^N \cap R)^d \geq |G| \cdot \operatorname{Fvol}(R, B).$$

Thus the left-hand side of (3) is at least $|G| \cdot \mathrm{Fvol}(R, B)$.

Conversely, let $\mathfrak{a}$ be an $\mathfrak{m}_R$-primary ideal. Then $\mathfrak{a} \neq 0$ since $d \geq 1$. Since any $g \in G$ fixes R pointwise and $g(S) = S$ by (1), we have $g(\mathfrak{a}S) = \mathfrak{a}S$ for any $g \in G$. Let Q be a prime ideal of S containing $\mathfrak{a}S$. Then $Q \cap R$ is a prime ideal containing $\mathfrak{a}$, so $Q \cap R = \mathfrak{m}_R$. Thus Q is a maximal ideal of S [HS06, Lemma 2.1.7], so $Q = \mathfrak{m}_S$. Since $\mathfrak{a}S \subset \mathfrak{m}_S$, the ideal $\mathfrak{a}S$ is $\mathfrak{m}_S$-primary. By (1) and [HS06, Theorem 11.2.7], we have

$$e_S(\mathfrak{a}S)\,\mathrm{fpt}(S, B_S; \mathfrak{a}S)^d = |G| \cdot e_R(\mathfrak{a})\,\mathrm{fpt}(R, B; \mathfrak{a})^d.$$

Taking the infimum over $\mathfrak{a}$, we see that the left-hand side of (3) is at most $|G| \cdot \mathrm{Fvol}(R, B)$, and (3) follows. □

The characteristic-zero counterpart of Proposition 5.5(3) is [LX17, Theorem 2.7(1)].

Proof of Theorem 5.2. Since S is complete, S is excellent [Gro65, (7.8.3)(iii)]. Moreover, S is F-finite [HS06, Theorem 4.3.3], and S is a normal local domain of characteristic p with $\dim S = d \geq 1$. By Definition 2.1, $B_S \geq 0$ since $B \geq 0$. By Proposition 5.5(1), (S, B_S) is strongly F-regular. Thus S, $\mathfrak{m}_S$, d, and B_S satisfy the conditions of Set-up 3.3. By Proposition 5.5(1) again, G is a finite group of ring automorphisms of S such that $g(B_S) = B_S$ for any $g \in G$. By Theorem 4.1(1) and Proposition 5.5(3), we have

$$\mathrm{Fvol}(S, B_S) = \inf_{\mathfrak{b}} e_S(\mathfrak{b})\,\mathrm{fpt}(S, B_S; \mathfrak{b})^d = |G| \cdot \mathrm{Fvol}(R, B),$$

where $\mathfrak{b}$ runs over the $\mathfrak{m}_S$-primary ideals of S such that $g(\mathfrak{b}) = \mathfrak{b}$ for any $g \in G$. □

6. Completion and Galois covers

In this section, we reduce questions about local rings essentially of finite type over k to the complete Galois case of Set-up 5.1. For a module-finite inclusion $R \subset S$ of normal local domains, the integral closure of R in a Galois closure of $\mathrm{Frac}\, S$ over $\mathrm{Frac}\, R$ is in general only semilocal, while Theorem 5.2 concerns Galois inclusions of local rings. Thus we first pass to the completions (Proposition 6.2).

6.1. **Completion.** In this subsection, we study the behavior of $\mathbb{Q}$-divisors, big test ideals, F-pure thresholds, and F-volumes under completion. In the case without boundary, big test ideals commute with completion (cf. [HT04, Proposition 3.2]), and for a boundary B, the new input is that the formation of C_e^B commutes with completion (Definition-Proposition 6.1(3)).

Definition-Proposition 6.1. Let $(R, \mathfrak{m}_R)$ be an excellent normal local domain with completion $\widehat{R}$. For a $\mathbb{Q}$-divisor $D = \sum_P d_P P$ on $\mathrm{Spec}\, R$, let

$$D_{\widehat{R}} := \sum_Q d_{Q \cap R} Q,$$

where Q runs over the height-one primes of $\widehat{R}$ such that $Q \cap R$ has height one. Let $B = \sum_P b_P P$ be a $\mathbb{Q}$-divisor on $\mathrm{Spec}\, R$. Then the following hold.

(1) For any height-one prime Q of $\widehat{R}$, the prime $Q \cap R$ is either zero or of height one, $\widehat{R}$ is a normal local domain, $D_{\widehat{R}}$ is a $\mathbb{Q}$-divisor on $\mathrm{Spec}\, \widehat{R}$ for any $\mathbb{Q}$-divisor D on $\mathrm{Spec}\, R$, and $D_{\widehat{R}} \geq 0$ if $D \geq 0$.

(2) For any real number $a \geq 0$, we have $\lceil aB \rceil_{\widehat{R}} = \lceil aB_{\widehat{R}} \rceil$ and

$$R(\lceil aB \rceil) \otimes_R \widehat{R} = \widehat{R}(\lceil aB_{\widehat{R}} \rceil)$$

as submodules of $\mathrm{Frac}\, \widehat{R}$.

(3) If R is F-finite of characteristic p and $B \geq 0$, then $\widehat{R}$ is F-finite, $\widehat{R} \otimes_R F_*^e R = F_*^e \widehat{R}$, and $C_e^B \otimes_R \widehat{R} = C_e^{B_{\widehat{R}}}$ as submodules of $\mathrm{Hom}_{\widehat{R}}(F_*^e \widehat{R}, \widehat{R})$ for any integer $e \geq 0$.

Proof. We prove (1). By [Gro65, (7.8.3)(v)], the map $R \to \widehat{R}$ is regular and $\widehat{R}$ is normal. Since $\widehat{R}$ is a local ring [Har77, Chapter I, Theorem 5.4A], it is a normal local domain. Let Q be a height-one prime of $\widehat{R}$. Since $\widehat{R}$ is flat over R [Har77, Chapter III, Example 9.1.2], going-down [HS06, Proposition B.1.2] implies that $\operatorname{ht}(Q \cap R) \leq \operatorname{ht} Q = 1$. Thus $Q \cap R$ is either zero or of height one. The sum defining $D_{\widehat{R}}$ is finite: if $f \in R$ is a non-zero element contained in all height-one primes P of R with $d_P \neq 0$, then any height-one prime Q of $\widehat{R}$ with $\operatorname{ht}(Q \cap R) = 1$ and $d_{Q \cap R} \neq 0$ is one of the finitely many minimal primes of $f\widehat{R}$. Thus $D_{\widehat{R}}$ is a $\mathbb{Q}$-divisor. The coefficients of $D_{\widehat{R}}$ are coefficients of D, so $D_{\widehat{R}} \geq 0$ if $D \geq 0$.

Next, let Q be a height-one prime of $\widehat{R}$ such that $P := Q \cap R$ has height one, and let t_P be a uniformizer of the discrete valuation ring R_P. We show that

$$\operatorname{ord}_Q(t_P) = 1. \tag{6.1}$$

Since $R \to \widehat{R}$ is regular, the ring $\widehat{R}_Q/P\widehat{R}_Q$, which is a localization of the fiber $\widehat{R} \otimes_R (R_P/PR_P)$, is a regular local ring. Moreover, $P\widehat{R}_Q = t_P\widehat{R}_Q$, and t_P is a non-zero element of the maximal ideal of the one-dimensional local domain $\widehat{R}_Q$, so $\dim \widehat{R}_Q/P\widehat{R}_Q = 0$. A regular local ring of dimension zero is a field. Thus t_P generates the maximal ideal of $\widehat{R}_Q$, which proves (6.1).

We prove (2). Let $a \geq 0$ be a real number. At any height-one prime Q of $\widehat{R}$, both $\lceil aB \rceil_{\widehat{R}}$ and $\lceil aB_{\widehat{R}} \rceil$ have coefficient $\lceil ab_{Q \cap R} \rceil$ if $\operatorname{ht}(Q \cap R) = 1$, and coefficient zero if $Q \cap R = 0$. Thus $\lceil aB \rceil_{\widehat{R}} = \lceil aB_{\widehat{R}} \rceil$. It remains to show that

$$R(D) \otimes_R \widehat{R} = \widehat{R}(D_{\widehat{R}}) \tag{6.2}$$

as submodules of $\operatorname{Frac} \widehat{R}$ for any Weil divisor $D = \sum_P d_P P$ on $\operatorname{Spec} R$. Since $\widehat{R}$ is flat over R, the map $R(D) \otimes_R \widehat{R} \to \operatorname{Frac} R \otimes_R \widehat{R}$ is injective, and $\operatorname{Frac} R \otimes_R \widehat{R}$ is a localization of $\widehat{R}$. Thus $R(D) \otimes_R \widehat{R}$ is an $\widehat{R}$-submodule of $\operatorname{Frac} \widehat{R}$. The R-module $R(D)$ is finitely generated, and it is reflexive by [Har80, Proposition 1.6], as membership in $R(D)$ is determined at the height-one primes of R. For any finitely generated R-module M, flat base change [Sta, Tag 087R] implies that

$$\operatorname{Hom}_R(M, R) \otimes_R \widehat{R} = \operatorname{Hom}_{\widehat{R}}(M \otimes_R \widehat{R}, \widehat{R}). \tag{6.3}$$

Applying (6.3) for $M = R(D)$ and for $M = \operatorname{Hom}_R(R(D), R)$, we see that $R(D) \otimes_R \widehat{R}$ is a reflexive $\widehat{R}$-module.

Now let Q be a height-one prime of $\widehat{R}$. If $P := Q \cap R$ has height one, then $R(D)_P = t_P^{-d_P} R_P$, where t_P is a uniformizer of R_P, so

$$\big(R(D) \otimes_R \widehat{R}\big)_Q = R(D)_P \otimes_{R_P} \widehat{R}_Q = t_P^{-d_P} \widehat{R}_Q = \widehat{R}(D_{\widehat{R}})_Q$$

by (6.1), as $D_{\widehat{R}}$ has coefficient d_P at Q. If $Q \cap R = 0$, then any non-zero element of R is a unit in $\widehat{R}_Q$, so

$$\big(R(D) \otimes_R \widehat{R}\big)_Q = \big(R(D) \otimes_R \operatorname{Frac} R\big) \otimes_{\operatorname{Frac} R} \widehat{R}_Q = \widehat{R}_Q = \widehat{R}(D_{\widehat{R}})_Q,$$

as $D_{\widehat{R}}$ has coefficient zero at Q. By [Har80, Proposition 1.6] and [Har77, Chapter II, Proposition 6.3A], any finitely generated reflexive $\widehat{R}$-submodule M' of $\operatorname{Frac} \widehat{R}$ satisfies $M' = \bigcap_Q M'_Q$, where Q runs over the height-one primes of $\widehat{R}$. By definition, $\widehat{R}(D_{\widehat{R}}) = \bigcap_Q \widehat{R}(D_{\widehat{R}})_Q$. Thus (6.2) holds. Taking $D = \lceil aB \rceil$ and using $\lceil aB \rceil_{\widehat{R}} = \lceil aB_{\widehat{R}} \rceil$ proves (2).

We prove (3). Let $e \geq 0$ be an integer and $q = p^e$. Since R is F-finite, $F^e_* M$ is a finitely generated R-module for any finitely generated R-module M. We show that

$$\widehat{R} \otimes_R F^e_* M = F^e_*(M \otimes_R \widehat{R}), \tag{6.4}$$

where $y \otimes F^e_* x$ corresponds to $F^e_*\big(y^q(x \otimes 1)\big)$ for $y \in \widehat{R}$ and $x \in M$. By [Har77, Chapter II, Theorem 9.3A], the left-hand side is the $\mathfrak{m}_R$-adic completion of $F^e_* M$, and $M \otimes_R \widehat{R}$ is the $\mathfrak{m}_R$-adic completion of M. For any integer $n \geq 1$, we have $\mathfrak{m}_R^n \cdot F^e_* M = F^e_*\big((\mathfrak{m}_R^n)^{[q]} M\big)$. Since the

ideal $(\mathfrak{m}_R^n)^{[q]}$ is $\mathfrak{m}_R$-primary, there exists an integer $n' \geq 1$ with $\mathfrak{m}_R^{n'} \subset (\mathfrak{m}_R^n)^{[q]} \subset \mathfrak{m}_R^n$. Thus the subgroups $(\mathfrak{m}_R^n)^{[q]}M$, $n \geq 1$, of M define the $\mathfrak{m}_R$-adic topology on M, and (6.4) follows. For $M = R$, (6.4) reads $\widehat{R} \otimes_R F_*^e R = F_*^e \widehat{R}$. For $e = 1$, it follows that $F_*^1 \widehat{R}$ is a finitely generated $\widehat{R}$-module, that is, $\widehat{R}$ is F-finite.

Let $D := \lceil (q-1)B \rceil$, and let

$$\rho : \mathrm{Hom}_R(F_*^e R(D), R) \to \mathrm{Hom}_R(F_*^e R, R)$$

be the restriction map, so that C_e^B is the image of ρ. By [Sta, Tag 087R], (6.4) for $M = R(D)$ and $M = R$, and (2) for $a = q-1$, we have

$$\begin{aligned} \mathrm{Hom}_R(F_*^e R(D), R) \otimes_R \widehat{R} &= \mathrm{Hom}_{\widehat{R}}\big(F_*^e \widehat{R}(\lceil (q-1)B_{\widehat{R}} \rceil), \widehat{R}\big), \\ \mathrm{Hom}_R(F_*^e R, R) \otimes_R \widehat{R} &= \mathrm{Hom}_{\widehat{R}}(F_*^e \widehat{R}, \widehat{R}). \end{aligned}$$

Since (6.4) and the isomorphism of [Sta, Tag 087R] are natural in M, the map $\rho \otimes_R \widehat{R}$ is the restriction map along $F_*^e \widehat{R} \subset F_*^e \widehat{R}(\lceil (q-1)B_{\widehat{R}} \rceil)$. Since $\widehat{R}$ is flat over R, the image of $\rho \otimes_R \widehat{R}$ is $C_e^B \otimes_R \widehat{R}$. Thus $C_e^B \otimes_R \widehat{R} = C_e^{B_{\widehat{R}}}$, where $B_{\widehat{R}} \geq 0$ by (1). □

Since $R \to \widehat{R}$ is regular, the factor $\mathrm{ord}_Q(t_{Q \cap R})$ of Definition 2.1 is equal to 1 by (6.1) for any height-one prime Q of $\widehat{R}$ such that $Q \cap R$ has height one, so it is omitted from $B_{\widehat{R}}$. The only difference from Definition 2.1 is that $R \to \widehat{R}$ is not module-finite: $\widehat{R}$ can have height-one primes Q such that $Q \cap R$ is zero, which do not appear in $B_{\widehat{R}}$, whereas in Definition 2.1, the prime $Q \cap R$ has height one for any height-one prime Q of S.

Since k is perfect, a local ring essentially of finite type over k is F-finite (see [ST10, Section 2.1]). Thus Definition 2.2 and Definition-Theorem 2.3 apply to R and S in Proposition 6.2 and Construction-Proposition 6.3, and, as R and S are excellent [Gro65, (7.8.3)(ii) and (iii)], also to $\widehat{R}$ and $\widehat{S}$ by Definition-Proposition 6.1(1) and (3).

Proposition 6.2. *Let $(R, \mathfrak{m}_R)$ be a normal local domain essentially of finite type over k with $R/\mathfrak{m}_R = k$ and $\dim R = d \geq 1$, and let $B \geq 0$ be a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$. Then the following hold.*

(1) For any non-zero ideal $\mathfrak{a} \subset R$ and any real number $t \geq 0$,

$$\tau_b(\widehat{R}, B_{\widehat{R}}, (\mathfrak{a}\widehat{R})^t) = \tau_b(R, B, \mathfrak{a}^t)\widehat{R}.$$

In particular, $(R, B, \mathfrak{a}^t)$ is strongly F-regular if and only if $(\widehat{R}, B_{\widehat{R}}, (\mathfrak{a}\widehat{R})^t)$ is strongly F-regular.

(2) Any $\mathfrak{m}_{\widehat{R}}$-primary ideal $\mathfrak{b}$ of $\widehat{R}$ satisfies $\mathfrak{b} = (\mathfrak{b} \cap R)\widehat{R}$, the ideal $\mathfrak{b} \cap R$ is $\mathfrak{m}_R$-primary, and $e_R(\mathfrak{b} \cap R) = e_{\widehat{R}}(\mathfrak{b})$.

(3) If (R, B) is strongly F-regular, then $(\widehat{R}, B_{\widehat{R}})$ is strongly F-regular, and

$$\mathrm{Fvol}(R, B) = \mathrm{Fvol}(\widehat{R}, B_{\widehat{R}}), \quad \mathrm{fpt}(R, B; \mathfrak{a}) = \mathrm{fpt}(\widehat{R}, B_{\widehat{R}}; \mathfrak{a}\widehat{R})$$

for any non-zero ideal $\mathfrak{a} \subset R$.

Proof. Since R is essentially of finite type over k, it is excellent [Gro65, (7.8.3)(ii) and (iii)]. By Definition-Proposition 6.1, $\widehat{R}$ is an F-finite normal local domain and $B_{\widehat{R}} \geq 0$ is a $\mathbb{Q}$-divisor on $\operatorname{Spec} \widehat{R}$. Moreover, $\mathfrak{m}_R\widehat{R}$ is the maximal ideal of $\widehat{R}$, and $\dim \widehat{R} = d$ [Har77, Chapter I, Theorem 5.4A].

We prove (1). Let $\mathfrak{a} \subset R$ be a non-zero ideal and $t \geq 0$ a real number. Fix an integer $e_0 \geq 1$ and let $q_0 = p^{e_0}$. By [Sta, Tag 087R],

$$\mathrm{Hom}_R\big(F_*^{e_0} R(\lceil (q_0 - 1)B \rceil), R\big) \otimes_R \mathrm{Frac}\, R = \mathrm{Hom}_{\mathrm{Frac}\, R}(F_*^{e_0} \mathrm{Frac}\, R, \mathrm{Frac}\, R),$$

which is a one-dimensional $F_*^{e_0}\operatorname{Frac} R$-vector space. Thus there exists a non-zero R-linear map $\widetilde{\psi} : F_*^{e_0}R(\lceil (q_0-1)B\rceil) \to R$, and the restriction $\psi \in C_{e_0}^B$ of $\widetilde{\psi}$ to $F_*^{e_0}R$ is non-zero: for any $z \in R(\lceil (q_0-1)B\rceil)$ and any non-zero $y \in R$ with $yz \in R$, we have $y^{q_0}z \in R$ and

$$\widetilde{\psi}(F_*^{e_0}(y^{q_0}z)) = y\widetilde{\psi}(F_*^{e_0}z),$$

and R is a domain. Moreover, the $F_*^{e_0}R$-linear map $F_*^{e_0}R \to \operatorname{Hom}_R(F_*^{e_0}R, R)$ sending 1 to ψ becomes surjective after tensoring with $\operatorname{Frac} R$. Let $u_1 \in \mathfrak{a}$ be a non-zero element. Since the regular locus of $\operatorname{Spec} R$ is open [Gro65, (7.8.3)(iv)] and contains the generic point, there exists a non-zero $u_2 \in R$ such that R_{u_2} is regular. Let $u_3 \in R$ be a non-zero element contained in all height-one primes P of R with $b_P \neq 0$. Let $u_4 \in R$ be a non-zero element annihilating the cokernel of $F_*^{e_0}R \to \operatorname{Hom}_R(F_*^{e_0}R, R)$. Let $u := u_1u_2u_3u_4 \in \mathfrak{a}$. Then R_u is regular, B restricted to $\operatorname{Spec} R_u$ is zero, and ψ generates $\operatorname{Hom}_{R_u}(F_*^{e_0}R_u, R_u)$ as an $F_*^{e_0}R_u$-module.

Let $\widehat{\psi} \in C_{e_0}^{B_{\widehat{R}}}$ be the image of $\psi \otimes 1$ under $C_{e_0}^B \otimes_R \widehat{R} = C_{e_0}^{B_{\widehat{R}}}$ (Definition-Proposition 6.1(3)). The restriction of $\widehat{\psi}$ to $F_*^{e_0}R$ is ψ, so $\widehat{\psi} \neq 0$. By [Gro65, (7.8.3)(v)], the regular locus of $\operatorname{Spec} \widehat{R}$ is the inverse image of the regular locus of $\operatorname{Spec} R$ under $\operatorname{Spec} \widehat{R} \to \operatorname{Spec} R$, so $\widehat{R}_u$ is regular. Any height-one prime Q of $\widehat{R}$ with $\operatorname{ht}(Q \cap R) = 1$ and $b_{Q\cap R} \neq 0$ contains u, so $B_{\widehat{R}}$ restricted to $\operatorname{Spec} \widehat{R}_u$ is zero. By [Sta, Tag 087R] and $\widehat{R} \otimes_R F_*^{e_0}R = F_*^{e_0}\widehat{R}$ (Definition-Proposition 6.1(3)), tensoring the surjection $F_*^{e_0}R_u \to \operatorname{Hom}_{R_u}(F_*^{e_0}R_u, R_u)$ sending 1 to ψ with $\widehat{R}$ over R yields the surjection $F_*^{e_0}\widehat{R}_u \to \operatorname{Hom}_{\widehat{R}_u}(F_*^{e_0}\widehat{R}_u, \widehat{R}_u)$ sending 1 to $\widehat{\psi}$.

Since R is essentially of finite type over k and $\widehat{R}$ is complete, the F-finite local rings R and $\widehat{R}$ have dualizing complexes [Har66, Chapter V, Section 10 and Corollary 2.3], and $\operatorname{Spec} R$ and $\operatorname{Spec} \widehat{R}$ satisfy condition (!) of [ST10, Section 2.5], as spectra of local rings. By [ST10, Lemma 6.5 and Remark 6.7], there exist positive integers n_1 and n_2 such that u^{n_1} satisfies the conclusion of [ST10, Lemma 6.5] for $(R, B, \mathfrak{a}^t)$ and ψ, and u^{n_2} satisfies the conclusion of [ST10, Lemma 6.5] for $(\widehat{R}, B_{\widehat{R}}, (\mathfrak{a}\widehat{R})^t)$ and $\widehat{\psi}$. That is, for any non-zero $w \in R$ and any non-zero $w' \in \widehat{R}$, there exist integers $n, n' \geq 1$ such that

$$u^{n_1} \in \psi_n\big(F_*^{ne_0}(w\mathfrak{a}^{\lceil t(p^{ne_0}-1)\rceil})\big), \quad u^{n_2} \in \widehat{\psi}_{n'}\big(F_*^{n'e_0}(w'(\mathfrak{a}\widehat{R})^{\lceil t(p^{n'e_0}-1)\rceil})\big),$$

where $\psi_n := \psi \circ F_*^{e_0}\psi \circ \cdots \circ F_*^{(n-1)e_0}\psi$ and $\widehat{\psi}_{n'} := \widehat{\psi} \circ F_*^{e_0}\widehat{\psi} \circ \cdots \circ F_*^{(n'-1)e_0}\widehat{\psi}$. By [ST10, Remark 6.6], both conditions hold for $c := u^{\max\{n_1,n_2\}}$ in place of u^{n_1} and u^{n_2}, which is the hypothesis on c in [ST10, Proposition 6.9] for R with ψ and for $\widehat{R}$ with $\widehat{\psi}$. For any integer $e \geq 0$, let $q = p^e$ and $\mathfrak{b}_e := c\mathfrak{a}^{\lceil t(q-1)\rceil}$. Then $c(\mathfrak{a}\widehat{R})^{\lceil t(q-1)\rceil} = \mathfrak{b}_e\widehat{R}$, and [ST10, Proposition 6.9] for R and for $\widehat{R}$ implies that

$$\begin{aligned} \tau_b(R, B, \mathfrak{a}^t) &= \sum_{e\geq 0}\sum_{\varphi\in C_e^B} \varphi(F_*^e\mathfrak{b}_e), \\ \tau_b(\widehat{R}, B_{\widehat{R}}, (\mathfrak{a}\widehat{R})^t) &= \sum_{e\geq 0}\sum_{\varphi\in C_e^{B_{\widehat{R}}}} \varphi\big(F_*^e(\mathfrak{b}_e\widehat{R})\big). \end{aligned} \tag{6.5}$$

Fix an integer $e \geq 0$ and let $q = p^e$. Since $\widehat{R}$ is flat over R, the inclusion $F_*^e\mathfrak{b}_e \subset F_*^eR$ induces an injection

$$\widehat{R} \otimes_R F_*^e\mathfrak{b}_e \to \widehat{R} \otimes_R F_*^eR = F_*^e\widehat{R},$$

whose image consists of the elements $F_*^e\big(\sum_j y_j^q x_j\big)$ with $y_j \in \widehat{R}$ and $x_j \in \mathfrak{b}_e$. By $\widehat{R} \otimes_R F_*^eR = F_*^e\widehat{R}$, any element of $\widehat{R}$ is of the form $\sum_j y_j^q x_j$ with $y_j \in \widehat{R}$ and $x_j \in R$. Thus the image is $F_*^e(\mathfrak{b}_e\widehat{R})$, that is,

$$\widehat{R} \otimes_R F_*^e\mathfrak{b}_e = F_*^e(\mathfrak{b}_e\widehat{R}).$$

The ideal $\sum_{\varphi\in C_e^B} \varphi(F_*^e\mathfrak{b}_e)$ is the image of the evaluation map $C_e^B \otimes_R F_*^e\mathfrak{b}_e \to R$, which sends $\varphi \otimes F_*^e x$ to $\varphi(F_*^e x)$. By Definition-Proposition 6.1(3) and $\widehat{R} \otimes_R F_*^e\mathfrak{b}_e = F_*^e(\mathfrak{b}_e\widehat{R})$, the base change

of the evaluation map along $R \to \widehat{R}$ is the evaluation map $C_e^{B_{\widehat{R}}} \otimes_{\widehat{R}} F_*^e(\mathfrak{b}_e\widehat{R}) \to \widehat{R}$. Since $\widehat{R}$ is flat over R, base change preserves images, and

$$\Big(\sum_{\varphi \in C_e^B} \varphi(F_*^e \mathfrak{b}_e)\Big)\widehat{R} = \sum_{\varphi \in C_e^{B_{\widehat{R}}}} \varphi\big(F_*^e(\mathfrak{b}_e\widehat{R})\big).$$

Summing over $e \geq 0$, we obtain $\tau_b(\widehat{R}, B_{\widehat{R}}, (\mathfrak{a}\widehat{R})^t) = \tau_b(R, B, \mathfrak{a}^t)\widehat{R}$ by (6.5).

By Definition-Theorem 2.3(1), $(R, B, \mathfrak{a}^t)$ is strongly F-regular if and only if $\tau_b(R, B, \mathfrak{a}^t) = R$, and similarly for $(\widehat{R}, B_{\widehat{R}}, (\mathfrak{a}\widehat{R})^t)$. If $\tau_b(R, B, \mathfrak{a}^t) \neq R$, then $\tau_b(R, B, \mathfrak{a}^t)\widehat{R} \subset \mathfrak{m}_R\widehat{R} \neq \widehat{R}$. Thus $(R, B, \mathfrak{a}^t)$ is strongly F-regular if and only if $(\widehat{R}, B_{\widehat{R}}, (\mathfrak{a}\widehat{R})^t)$ is strongly F-regular. This proves (1) (cf. [HT04, Proposition 3.2] for the case $B = 0$).

We prove (2). By [Har77, Chapter I, Theorem 5.4A and Chapter II, Theorem 9.3A], we have $\mathfrak{m}_{\widehat{R}}^n = \mathfrak{m}_R^n\widehat{R}$ and $R/\mathfrak{m}_R^n = \widehat{R}/\mathfrak{m}_R^n\widehat{R}$ for any integer $n \geq 1$. Since primary ideals contain powers of the maximal ideals, it follows that the maps sending $\mathfrak{a}$ to $\mathfrak{a}\widehat{R}$ and $\mathfrak{b}$ to $\mathfrak{b} \cap R$ are mutually inverse bijections between the $\mathfrak{m}_R$-primary ideals of R and the $\mathfrak{m}_{\widehat{R}}$-primary ideals of $\widehat{R}$, and $R/\mathfrak{a} = \widehat{R}/\mathfrak{a}\widehat{R}$ for any $\mathfrak{m}_R$-primary ideal $\mathfrak{a}$ of R. Let $\mathfrak{a}$ be an $\mathfrak{m}_R$-primary ideal of R. For any integer $m \geq 1$, the ideal $\mathfrak{a}^m$ is $\mathfrak{m}_R$-primary and $(\mathfrak{a}\widehat{R})^m = \mathfrak{a}^m\widehat{R}$, so $R/\mathfrak{a}^m = \widehat{R}/(\mathfrak{a}\widehat{R})^m$. Since $R \to \widehat{R}/(\mathfrak{a}\widehat{R})^m$ is surjective, the R-submodules and the $\widehat{R}$-submodules of $\widehat{R}/(\mathfrak{a}\widehat{R})^m$ coincide, so $\ell_R(R/\mathfrak{a}^m) = \ell_{\widehat{R}}(\widehat{R}/(\mathfrak{a}\widehat{R})^m)$. As $\dim \widehat{R} = \dim R = d$, it follows that

$$e_R(\mathfrak{a}) = e_{\widehat{R}}(\mathfrak{a}\widehat{R}). \tag{6.6}$$

For an $\mathfrak{m}_{\widehat{R}}$-primary ideal $\mathfrak{b}$ of $\widehat{R}$, taking $\mathfrak{a} = \mathfrak{b} \cap R$ proves (2).

We prove (3). Assume that (R, B) is strongly F-regular. By (1) for $\mathfrak{a} = R$ and $t = 1$, the pair $(\widehat{R}, B_{\widehat{R}})$ is strongly F-regular. Let $\mathfrak{a} \subset R$ be a non-zero ideal. By (1), for any real number $t \geq 0$, the triple $(R, B, \mathfrak{a}^t)$ is strongly F-regular if and only if $(\widehat{R}, B_{\widehat{R}}, (\mathfrak{a}\widehat{R})^t)$ is strongly F-regular. Thus $\operatorname{fpt}(R, B; \mathfrak{a}) = \operatorname{fpt}(\widehat{R}, B_{\widehat{R}}; \mathfrak{a}\widehat{R})$. Hence, by (6.6), we obtain

$$e_R(\mathfrak{a}) \operatorname{fpt}(R, B; \mathfrak{a})^d = e_{\widehat{R}}(\mathfrak{a}\widehat{R}) \operatorname{fpt}(\widehat{R}, B_{\widehat{R}}; \mathfrak{a}\widehat{R})^d$$

for any $\mathfrak{m}_R$-primary ideal $\mathfrak{a}$ of R. Since the map sending $\mathfrak{a}$ to $\mathfrak{a}\widehat{R}$ is a bijection from the $\mathfrak{m}_R$-primary ideals of R to the $\mathfrak{m}_{\widehat{R}}$-primary ideals of $\widehat{R}$, taking the infimum over $\mathfrak{a}$ implies that $\operatorname{Fvol}(R, B) = \operatorname{Fvol}(\widehat{R}, B_{\widehat{R}})$. □

6.2. **Galois covers.** In this subsection, for a module-finite inclusion $R \subset S$ of normal local domains such that $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale in codimension one, we construct a complete local Galois cover of $\widehat{R}$ that contains $\widehat{S}$ and is étale in codimension one over $\widehat{R}$ and over $\widehat{S}$, using that the integral closure of $\widehat{R}$ in a finite Galois extension of $\operatorname{Frac}\widehat{R}$ is local, as $\widehat{R}$ is complete. A similar construction appears in [CST18, Section 2.4]. In Construction-Proposition 6.3, item (3) relates the orders of the Galois groups of the cover over $\widehat{R}$ and over $\widehat{S}$, and item (4) shows that the boundaries on the cover pulled back from $\widehat{R}$ and from $\widehat{S}$ coincide, which is used in the proof of Theorem 1.3.

Construction-Proposition 6.3. Let $(R, \mathfrak{m}_R) \subset (S, \mathfrak{m}_S)$ be a module-finite inclusion of normal local domains essentially of finite type over k, with $R/\mathfrak{m}_R = S/\mathfrak{m}_S = k$ and $\dim R = d \geq 1$, and let $B \geq 0$ be a $\mathbb{Q}$-divisor on $\operatorname{Spec} R$. Let $r := [\operatorname{Frac} S : \operatorname{Frac} R]$. Assume that $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale in codimension one and (R, B) is strongly F-regular. Then the following hold.

(1) $\widehat{R} \subset \widehat{S}$ is a module-finite inclusion of complete Noetherian normal local domains of dimension d with residue field k, $\operatorname{Spec}\widehat{S} \to \operatorname{Spec}\widehat{R}$ is étale in codimension one, the degree $[\operatorname{Frac}\widehat{S} : \operatorname{Frac}\widehat{R}]$ equals r, and $B_{\widehat{S}} := (B_S)_{\widehat{S}}$ is the divisorial pullback of $B_{\widehat{R}}$ to $\operatorname{Spec}\widehat{S}$.

(2) There exists a finite Galois extension M of $\operatorname{Frac}\widehat{R}$ containing $\operatorname{Frac}\widehat{S}$ such that the integral closure $\widetilde{S}$ of $\widehat{R}$ in M is a complete Noetherian normal local domain of dimension d with $\operatorname{Frac}\widetilde{S} = M$ and residue field k, module-finite over $\widehat{S}$, and both $\operatorname{Spec}\widetilde{S} \to \operatorname{Spec}\widehat{R}$ and $\operatorname{Spec}\widetilde{S} \to \operatorname{Spec}\widehat{S}$ are étale in codimension one.
(3) For $G := \operatorname{Gal}(M/\operatorname{Frac}\widehat{R})$ and $H := \operatorname{Gal}(M/\operatorname{Frac}\widehat{S})$, we have $\widetilde{S}^G = \widehat{R}$, $\widetilde{S}^H = \widehat{S}$, and $|G| = r|H|$.
(4) The divisorial pullback $B_{\widetilde{S}}$ of $B_{\widehat{R}}$ to $\operatorname{Spec}\widetilde{S}$ is the divisorial pullback of $B_{\widehat{S}}$ to $\operatorname{Spec}\widetilde{S}$.
(5) The pairs $(\widehat{R}, B_{\widehat{R}})$ and $(\widehat{S}, B_{\widehat{S}})$ are strongly F-regular.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

Since S is integral over R, we have $\dim S = \dim R = d$ [HS06, Theorem 2.2.5]. Thus R and S are normal local domains essentially of finite type over k of dimension d with residue field k, so their completions $\widehat{R}$ and $\widehat{S}$ are complete Noetherian normal local domains of dimension d with residue field k [Gro65, (7.8.3)(v)].

The ring $S/\mathfrak{m}_R S$ is a local ring with $\dim_k S/\mathfrak{m}_R S < +\infty$, hence an Artinian local ring. Thus there exists a positive integer n with $\mathfrak{m}_S^n \subset \mathfrak{m}_R S \subset \mathfrak{m}_S$, and the $\mathfrak{m}_R S$-adic and $\mathfrak{m}_S$-adic topologies on S coincide. Hence $S \otimes_R \widehat{R} = \widehat{S}$ by [Har77, Chapter II, Theorem 9.3A]. Since $\widehat{R}$ is flat over R [Har77, Chapter III, Example 9.1.2], the map $\widehat{R} = R \otimes_R \widehat{R} \to S \otimes_R \widehat{R} = \widehat{S}$ is injective. Thus $\widehat{R} \subset \widehat{S}$ is a module-finite inclusion. Since $\widehat{S}$ is a domain and a finitely generated $\widehat{R}$-module, the localization $\widehat{S} \otimes_{\widehat{R}} \operatorname{Frac}\widehat{R}$ of $\widehat{S}$ is a domain with $\dim_{\operatorname{Frac}\widehat{R}} \widehat{S} \otimes_{\widehat{R}} \operatorname{Frac}\widehat{R} < +\infty$, hence a field. Similarly, $S \otimes_R \operatorname{Frac} R$ is a field. Thus

$$\operatorname{Frac}\widehat{S} = \widehat{S} \otimes_{\widehat{R}} \operatorname{Frac}\widehat{R} = S \otimes_R \operatorname{Frac}\widehat{R} = \operatorname{Frac} S \otimes_{\operatorname{Frac} R} \operatorname{Frac}\widehat{R},$$

and $[\operatorname{Frac}\widehat{S} : \operatorname{Frac}\widehat{R}] = [\operatorname{Frac} S : \operatorname{Frac} R] = r$.

Let Q be a prime of $\widehat{S}$ of height at most one. Since $\widehat{S}$ is flat over S [Har77, Chapter III, Example 9.1.2], going-down [HS06, Proposition B.1.2] implies that $\operatorname{ht}(Q \cap S) \leq 1$. Let $U_S \subset \operatorname{Spec} S$ be the set of points at which $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale. Then U_S is open [Gro71, Exp. I, Proposition 4.5], and $Q \cap S \in U_S$ since $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale in codimension one. The open subset $U_S \times_{\operatorname{Spec} R} \operatorname{Spec}\widehat{R}$ of $\operatorname{Spec} S \times_{\operatorname{Spec} R} \operatorname{Spec}\widehat{R} = \operatorname{Spec}\widehat{S}$ contains Q, and it is étale over $\operatorname{Spec}\widehat{R}$ [Gro71, Exp. I, Proposition 4.6(iii)]. Thus $\operatorname{Spec}\widehat{S} \to \operatorname{Spec}\widehat{R}$ is étale in codimension one.

Finally, we show that $B_{\widehat{S}}$ is the divisorial pullback of $B_{\widehat{R}}$. Let Q be a height-one prime of $\widehat{S}$. By Definition 2.1, $Q \cap \widehat{R}$ has height one. By Definition-Proposition 6.1(1), each of $Q \cap S$ and $Q \cap R = (Q \cap \widehat{R}) \cap R$ is either zero or of height one. If $Q \cap S = 0$, then $Q \cap R = 0$. If $Q \cap S \neq 0$, then $Q \cap R = (Q \cap S) \cap R$ has height one by Definition 2.1. Since $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale, hence unramified, at $Q \cap S$, we have $\operatorname{ord}_{Q\cap S}(t_{Q\cap R}) = 1$ [Gro71, Exp. I, Proposition 3.1], where $t_{Q\cap R}$ is a uniformizer of $R_{Q\cap R}$. Thus the coefficient of $B_{\widehat{S}} = (B_S)_{\widehat{S}}$ at Q is $b_{Q\cap R}$ if $Q \cap R \neq 0$, and zero if $Q \cap R = 0$. Since $\operatorname{Spec}\widehat{S} \to \operatorname{Spec}\widehat{R}$ is étale, hence unramified, at Q, we have $\operatorname{ord}_Q(t_{Q\cap\widehat{R}}) = 1$ [Gro71, Exp. I, Proposition 3.1], where $t_{Q\cap\widehat{R}}$ is a uniformizer of $\widehat{R}_{Q\cap\widehat{R}}$. Thus the coefficient at Q of the divisorial pullback of $B_{\widehat{R}}$ is the coefficient of $B_{\widehat{R}}$ at $Q \cap \widehat{R}$, which is $b_{Q\cap R}$ if $Q \cap R \neq 0$, and zero if $Q \cap R = 0$. Hence $B_{\widehat{S}}$ is the divisorial pullback of $B_{\widehat{R}}$, and (1) holds.

Step 2. In this step, we construct M and $\widetilde{S}$, and we prove (2) and (3).

Let $\mathfrak{b} \subset \widehat{S}$ be an ideal such that $V(\mathfrak{b})$ is the set of points at which $\operatorname{Spec}\widehat{S} \to \operatorname{Spec}\widehat{R}$ is not étale [Gro71, Exp. I, Proposition 4.5]. By [HS06, Theorem 2.2.2] for the integral extension $\widehat{R}/(\mathfrak{b} \cap \widehat{R}) \subset \widehat{S}/\mathfrak{b}$, the image of $V(\mathfrak{b})$ in $\operatorname{Spec}\widehat{R}$ is the closed subset $V(\mathfrak{b} \cap \widehat{R})$. Let P be a prime of $\widehat{R}$ of height at most one, and Q a prime of $\widehat{S}$ with $Q \cap \widehat{R} = P$. By [HS06, Theorem 2.2.3], a strictly increasing chain of primes of $\widehat{S}$ contained in Q contracts to a strictly increasing chain of

primes of $\widehat{R}$ contained in P, so $\operatorname{ht} Q \leq \operatorname{ht} P \leq 1$, and $Q \notin V(\mathfrak{b})$ by (1). Thus the open subset

$$U := \operatorname{Spec} \widehat{R} \backslash V(\mathfrak{b} \cap \widehat{R})$$

contains all points of codimension at most one, and the morphism $U_{\widehat{S}} := \operatorname{Spec} \widehat{S} \times_{\operatorname{Spec} \widehat{R}} U \to U$ is finite and étale.

As non-empty open subsets of the integral schemes $\operatorname{Spec} \widehat{R}$ and $\operatorname{Spec} \widehat{S}$, the schemes U and $U_{\widehat{S}}$ are connected. By [Sta, Tag 03SF], there exist a finite étale Galois cover $W \to U$ and a U-morphism $W \to U_{\widehat{S}}$. By [Gro71, Exp. I, Proposition 10.1], W is integral, and the function field M of W is a finite separable extension of $\operatorname{Frac} \widehat{R}$. The U-morphism $W \to U_{\widehat{S}}$ is étale [Gro71, Exp. I, Corollaire 4.8], hence open, so it maps the generic point of W to the generic point of $U_{\widehat{S}}$. Thus we obtain an inclusion $\operatorname{Frac} \widehat{S} \subset M$ of extensions of $\operatorname{Frac} \widehat{R}$. Let $\operatorname{Aut}_U(W)$ be the group of U-automorphisms of W. Since W is integral and separated over U, any $\sigma \in \operatorname{Aut}_U(W)$ is determined by the automorphism of M induced by σ. Thus

$$|\operatorname{Aut}(M/\operatorname{Frac} \widehat{R})| \geq |\operatorname{Aut}_U(W)| = [M : \operatorname{Frac} \widehat{R}],$$

where the equality holds because $W \to U$ is a Galois cover of degree $[M : \operatorname{Frac} \widehat{R}]$. Hence M is a Galois extension of $\operatorname{Frac} \widehat{R}$.

Let $\widetilde{S}$ be the integral closure of $\widehat{R}$ in M. By [HS06, Theorem 4.3.4], $\widetilde{S}$ is a complete local ring which is module-finite over $\widehat{R}$. For any $z \in M$, there exists a non-zero $y \in \widehat{R}$ such that yz is integral over $\widehat{R}$, since z is algebraic over $\operatorname{Frac} \widehat{R}$. Thus $\operatorname{Frac} \widetilde{S} = M$, and $\widetilde{S}$ is a normal domain. Since $\widetilde{S}$ is integral over $\widehat{R}$, we have $\dim \widetilde{S} = d$ [HS06, Theorem 2.2.5]. The residue field of $\widetilde{S}$ is a finite extension of k, hence equal to k, as k is algebraically closed. Since $\widehat{S} \subset \operatorname{Frac} \widehat{S} \subset M$ and $\widehat{S}$ is integral over $\widehat{R}$, we have $\widehat{S} \subset \widetilde{S}$, and $\widetilde{S}$ is module-finite over $\widehat{S}$, hence Noetherian.

By [Gro71, Exp. I, Corollaire 10.2], W is isomorphic over U to the normalization of U in M, that is, to $U_{\widetilde{S}} := \operatorname{Spec} \widetilde{S} \times_{\operatorname{Spec} \widehat{R}} U$. Thus $U_{\widetilde{S}}$ is étale over U. Let Q be a prime of $\widetilde{S}$ of height at most one. Since $\widehat{R}$ is normal and $\widetilde{S}$ is a domain integral over $\widehat{R}$, going-down [HS06, Theorem 2.2.7] implies that $\operatorname{ht}(Q \cap \widehat{R}) \leq 1$. Thus $Q \cap \widehat{R} \in U$, so $Q \in U_{\widetilde{S}}$ and $Q \cap \widehat{S} \in U_{\widehat{S}}$. Hence $\operatorname{Spec} \widetilde{S} \to \operatorname{Spec} \widehat{R}$ is étale at Q. The U-morphism $U_{\widetilde{S}} \to U_{\widehat{S}}$ induced by $\widehat{S} \subset \widetilde{S}$ is étale [Gro71, Exp. I, Corollaire 4.8], as $U_{\widetilde{S}}$ and $U_{\widehat{S}}$ are étale over U. Hence $\operatorname{Spec} \widetilde{S} \to \operatorname{Spec} \widehat{S}$ is étale at Q. This proves (2).

Since M is Galois over $\operatorname{Frac} \widehat{R}$, it is Galois over $\operatorname{Frac} \widehat{S}$, and $H = \operatorname{Gal}(M/\operatorname{Frac} \widehat{S})$ is a subgroup of $G = \operatorname{Gal}(M/\operatorname{Frac} \widehat{R})$. Any $g \in G$ fixes $\widehat{R}$, so $g(\widetilde{S}) = \widetilde{S}$. The ring $\widetilde{S}^G$ consists of the elements of $M^G = \operatorname{Frac} \widehat{R}$ which are integral over $\widehat{R}$, so $\widetilde{S}^G = \widehat{R}$, as $\widehat{R}$ is normal. Similarly, $\widetilde{S}^H$ consists of the elements of $\operatorname{Frac} \widehat{S}$ which are integral over $\widehat{R}$, hence over $\widehat{S}$, so $\widetilde{S}^H = \widehat{S}$, as $\widehat{S}$ is normal. By (1),

$$|G| = [M : \operatorname{Frac} \widehat{R}] = [M : \operatorname{Frac} \widehat{S}][\operatorname{Frac} \widehat{S} : \operatorname{Frac} \widehat{R}] = r|H|.$$

This proves (3).

Step 3. In this step, we prove (4) and (5).

Let Q be a height-one prime of $\widetilde{S}$. By Definition 2.1, $Q \cap \widehat{S}$ and $Q \cap \widehat{R}$ have height one. By (1) and (2), $\operatorname{Spec} \widetilde{S} \to \operatorname{Spec} \widehat{R}$ and $\operatorname{Spec} \widetilde{S} \to \operatorname{Spec} \widehat{S}$ are étale, hence unramified, at Q, and $\operatorname{Spec} \widehat{S} \to \operatorname{Spec} \widehat{R}$ is étale, hence unramified, at $Q \cap \widehat{S}$. By [Gro71, Exp. I, Proposition 3.1],

$$\operatorname{ord}_Q(t_{Q \cap \widehat{R}}) = \operatorname{ord}_Q(t_{Q \cap \widehat{S}}) = \operatorname{ord}_{Q \cap \widehat{S}}(t_{Q \cap \widehat{R}}) = 1,$$

where $t_{Q \cap \widehat{R}}$ and $t_{Q \cap \widehat{S}}$ are uniformizers of $\widehat{R}_{Q \cap \widehat{R}}$ and $\widehat{S}_{Q \cap \widehat{S}}$. Since $(Q \cap \widehat{S}) \cap \widehat{R} = Q \cap \widehat{R}$, it follows from Definition 2.1 and (1) that the coefficient of $B_{\widetilde{S}}$ at Q, the coefficient at Q of the divisorial pullback of $B_{\widehat{S}}$ to $\operatorname{Spec} \widetilde{S}$, and the coefficient of $B_{\widehat{S}}$ at $Q \cap \widehat{S}$ are all equal to the coefficient of $B_{\widehat{R}}$ at $Q \cap \widehat{R}$. This proves (4).

Since (R, B) is strongly F-regular, R is strongly F-regular (Lemma 2.6(1)). By Theorem 2.7(3) for $R \subset S$, the pair (S, B_S) is strongly F-regular. Since $S/\mathfrak{m}_S = k$, $\dim S = d$ by **Step 1**, and $B_S \geq 0$, Proposition 6.2(3) for (R, B) and for (S, B_S) implies that $(\widehat{R}, B_{\widehat{R}})$ and $(\widehat{S}, B_{\widehat{S}})$ are strongly F-regular. This proves (5). □

7. Proofs of the main theorems

In this section, we prove Theorems 1.2, 1.3, and 1.4.

Proof of Theorem 1.4. Let $R := \mathcal{O}_{X,x}$. Then $\dim R = d$, and by Definition 2.4, it suffices to show that $\mathrm{Fvol}_{\mathrm{HK}}(R, B_x) = \mathrm{Fvol}(R, B_x)/d!$.

Any $\mathfrak{m}_R$-primary ideal $\mathfrak{a}$ of R satisfies $e_{\mathrm{HK},R}(\mathfrak{a}) \geq e_R(\mathfrak{a})/d!$ [WY01, Lemma 1.2]. Thus

$$e_{\mathrm{HK},R}(\mathfrak{a}) \operatorname{fpt}(R, B_x; \mathfrak{a})^d \geq \frac{e_R(\mathfrak{a}) \operatorname{fpt}(R, B_x; \mathfrak{a})^d}{d!} \geq \frac{\mathrm{Fvol}(R, B_x)}{d!}.$$

Taking the infimum over $\mathfrak{a}$, we get $\mathrm{Fvol}_{\mathrm{HK}}(R, B_x) \geq \mathrm{Fvol}(R, B_x)/d!$.

Conversely, fix an $\mathfrak{m}_R$-primary ideal $\mathfrak{a}$ of R. For any positive integer m, the ideal $\mathfrak{a}^m$ is $\mathfrak{m}_R$-primary, and $\operatorname{fpt}(R, B_x; \mathfrak{a}^m) = \operatorname{fpt}(R, B_x; \mathfrak{a})/m$ by Lemma 2.6(4). Thus

$$\mathrm{Fvol}_{\mathrm{HK}}(R, B_x) \leq e_{\mathrm{HK},R}(\mathfrak{a}^m) \operatorname{fpt}(R, B_x; \mathfrak{a}^m)^d = \frac{e_{\mathrm{HK},R}(\mathfrak{a}^m)}{m^d} \operatorname{fpt}(R, B_x; \mathfrak{a})^d.$$

Since $e_R(\mathfrak{a}^m) = m^d e_R(\mathfrak{a})$ [HS06, Proposition 11.2.9(1)], we have

$$\lim_{m \to +\infty} \frac{e_{\mathrm{HK},R}(\mathfrak{a}^m)}{m^d} = e_R(\mathfrak{a}) \cdot \lim_{m \to +\infty} \frac{e_{\mathrm{HK},R}(\mathfrak{a}^m)}{e_R(\mathfrak{a}^m)} = \frac{e_R(\mathfrak{a})}{d!},$$

where the last equality follows from [WY01, Theorem 1.1]. Letting $m \to +\infty$ and using that $\operatorname{fpt}(R, B_x; \mathfrak{a})$ is finite (Lemma 2.6(5)), we obtain

$$\mathrm{Fvol}_{\mathrm{HK}}(R, B_x) \leq \frac{e_R(\mathfrak{a}) \operatorname{fpt}(R, B_x; \mathfrak{a})^d}{d!}.$$

Taking the infimum over $\mathfrak{a}$, we get $\mathrm{Fvol}_{\mathrm{HK}}(R, B_x) \leq \mathrm{Fvol}(R, B_x)/d!$. □

Proof of Theorem 1.3. Let $R := \mathcal{O}_{X,x}$ and $S := \mathcal{O}_{Y,y}$. Since f is finite and surjective, $(f_*\mathcal{O}_Y)_x$ is a module-finite extension domain of R whose maximal ideals are the primes contracting to $\mathfrak{m}_R$ [HS06, Lemma 2.1.7], that is, the points of $f^{-1}(x) = \{y\}$. Hence $(f_*\mathcal{O}_Y)_x = S$. Thus $R \subset S$ is a module-finite inclusion of F-finite normal local domains essentially of finite type over k (see [ST10, Section 2.1] for F-finiteness), with $R/\mathfrak{m}_R = S/\mathfrak{m}_S = k$ and $\dim R = \dim S = d$. Moreover, $[\operatorname{Frac} S : \operatorname{Frac} R] = \deg(f)$, and $\operatorname{Spec} S \to \operatorname{Spec} R$ is étale in codimension one, as f is.

Let B_S be the divisorial pullback of B_x to $\operatorname{Spec} S$. The height-one primes of S correspond to the prime divisors of Y through y, so $B_S = (B_Y)_y$ by Definition 2.1. Since R is strongly F-regular (Lemma 2.6(1)), Theorem 2.7(3) shows that (S, B_S) is strongly F-regular, that is, $(Y \ni y, B_Y)$ is strongly F-regular.

We use the notation of Construction-Proposition 6.3 for $R \subset S$ and B_x, in which $B_{\widehat{R}}$ denotes the completion of B_x (Definition-Proposition 6.1) and $r = \deg(f)$. By Construction-Proposition 6.3(1)–(5), together with $k \subset R \subset \widehat{R}$ and the fact that $M/\operatorname{Frac} \widehat{S}$ is a Galois extension with group H (as $\operatorname{Frac} \widehat{S}$ is an intermediate field of the Galois extension $M/\operatorname{Frac} \widehat{R}$), the conditions of Set-up 5.1 hold with (R, S, G, B, B_S) replaced by $(\widehat{R}, \widetilde{S}, G, B_{\widehat{R}}, B_{\widetilde{S}})$, and also with (R, S, G, B, B_S) replaced by $(\widehat{S}, \widetilde{S}, H, B_{\widehat{S}}, B_{\widetilde{S}})$. By Theorem 5.2, applied in both cases, we get

$$|G| \cdot \mathrm{Fvol}(\widehat{R}, B_{\widehat{R}}) = \mathrm{Fvol}(\widetilde{S}, B_{\widetilde{S}}) = |H| \cdot \mathrm{Fvol}(\widehat{S}, B_{\widehat{S}}).$$

Since $\mathrm{Fvol}(\widehat{R}, B_{\widehat{R}})$ is finite by Lemma 2.6(5) and $|G| = r|H| = \deg(f)|H|$ by Construction-Proposition 6.3(3), dividing by $|H|$ shows that

$$\mathrm{Fvol}(\widehat{S}, B_{\widehat{S}}) = \deg(f) \cdot \mathrm{Fvol}(\widehat{R}, B_{\widehat{R}}).$$

By Proposition 6.2(3), applied to (R, B_x) and to (S, B_S), we have $\mathrm{Fvol}(R, B_x) = \mathrm{Fvol}(\widehat{R}, B_{\widehat{R}})$ and $\mathrm{Fvol}(S, B_S) = \mathrm{Fvol}(\widehat{S}, B_{\widehat{S}})$. Thus

$$\mathrm{Fvol}(Y \ni y, B_Y) = \mathrm{Fvol}(S, B_S) = \deg(f) \cdot \mathrm{Fvol}(R, B_x) = \deg(f) \cdot \mathrm{Fvol}(X \ni x, B).$$

Finally, Theorem 1.4, applied to $(X \ni x, B)$ and to $(Y \ni y, B_Y)$, implies

$$\begin{aligned}\mathrm{Fvol}_{\mathrm{HK}}(Y \ni y, B_Y) &= \frac{\mathrm{Fvol}(Y \ni y, B_Y)}{d!} = \frac{\deg(f) \cdot \mathrm{Fvol}(X \ni x, B)}{d!} \\ &= \deg(f) \cdot \mathrm{Fvol}_{\mathrm{HK}}(X \ni x, B).\end{aligned}$$

□

Proof of Theorem 1.2. Theorem 1.2 is the case $B = 0$ of Theorem 1.3, as $f^*0 = 0$. □

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